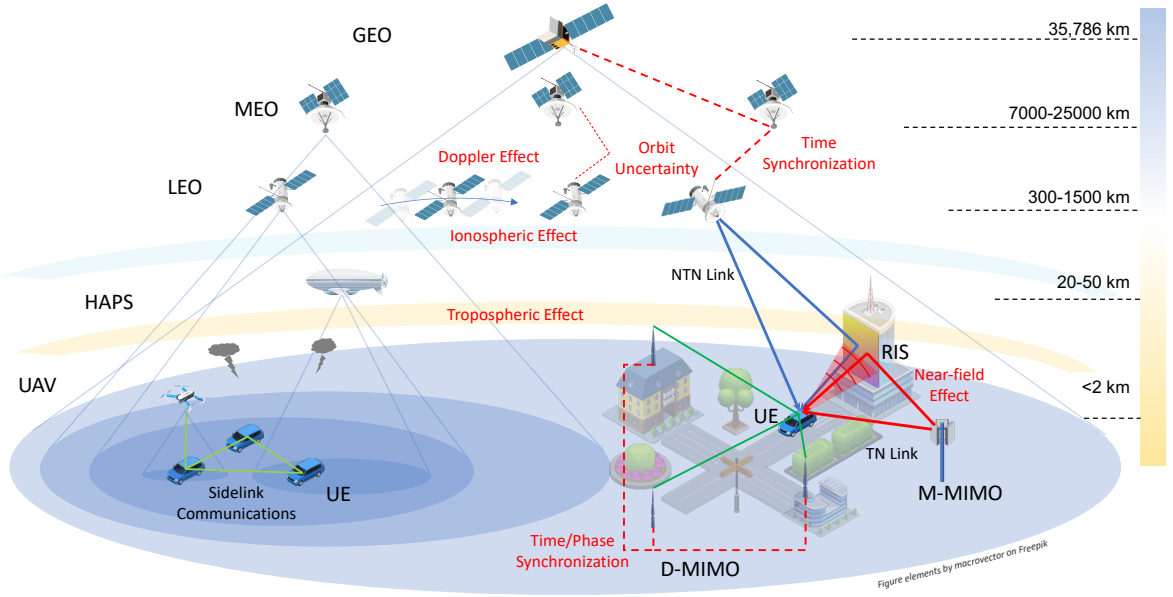
# Part III: NTN Characteristics and Localization Techniques

Dr. Pinjun Zheng

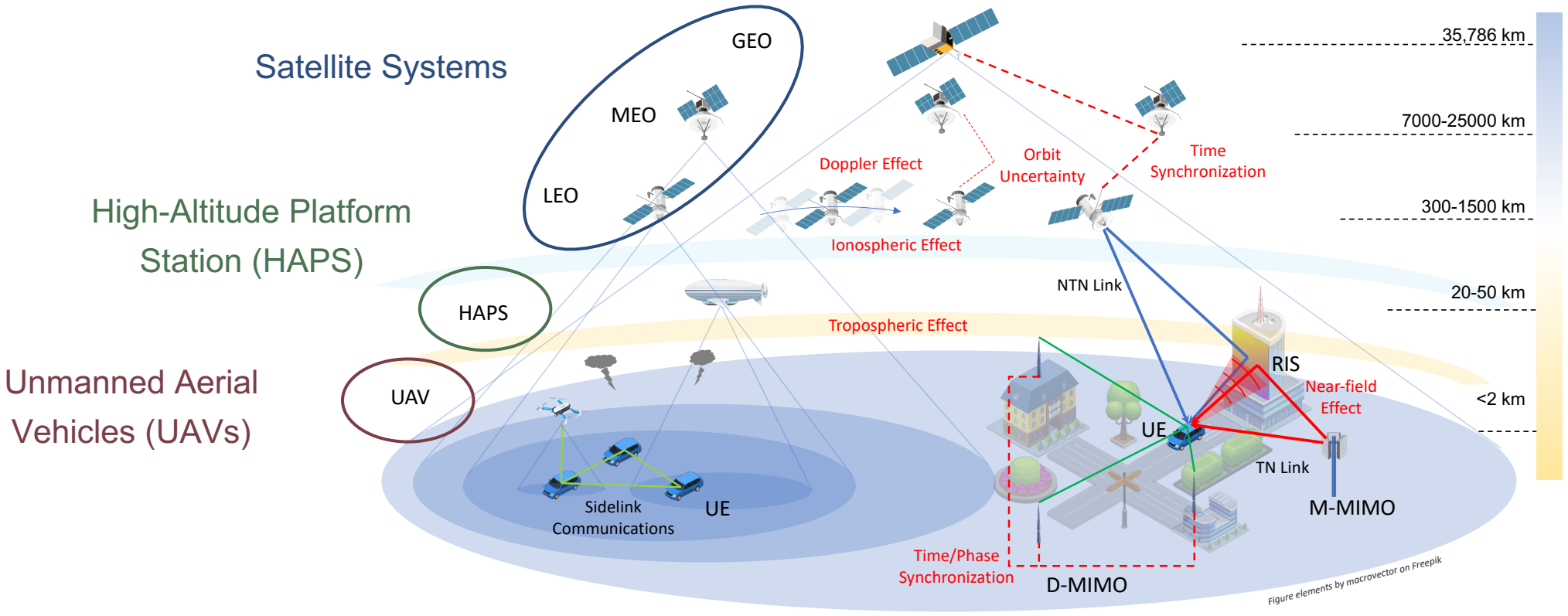
CHALMERS
UNIVERSITY OF TECHNOLOGY
UBC

# Part III: NTN Characteristics and Localization Techniques

- Non-Terrestrial Systems
- Localization Based on GNSS
- Localization Based on LEO Satellites
- Case Studies



# Introduction to Non-Terrestrial Systems




[1] S. Saleh et al., "Integrated 6G TN and NTN Localization: Challenges, Opportunities, and Advancements," in IEEE Communications Standards Magazine, vol. 9, no. 2, pp. 63-71, June 2025.

# Introduction to Non-Terrestrial Systems



## UAV

- **Height:** 0.1–5 km; localized coverage
- **Latency:** Very low
- **Flexibility:** Highly flexible, on-demand
- **Sensing:** Multimodal; RF, visual, LiDAR, radar, etc.
- **Typical use cases:** Emergency localization and UAV swarms



## HAPS

- **Height:** ~20 km; regional coverage
- **Latency:** Low–moderate
- **Flexibility:** Semi-stationary
- **Sensing:** Primarily RF-based, limited visual payload
- **Typical use cases:** Rural Coverage and disaster recovery



## Satellite

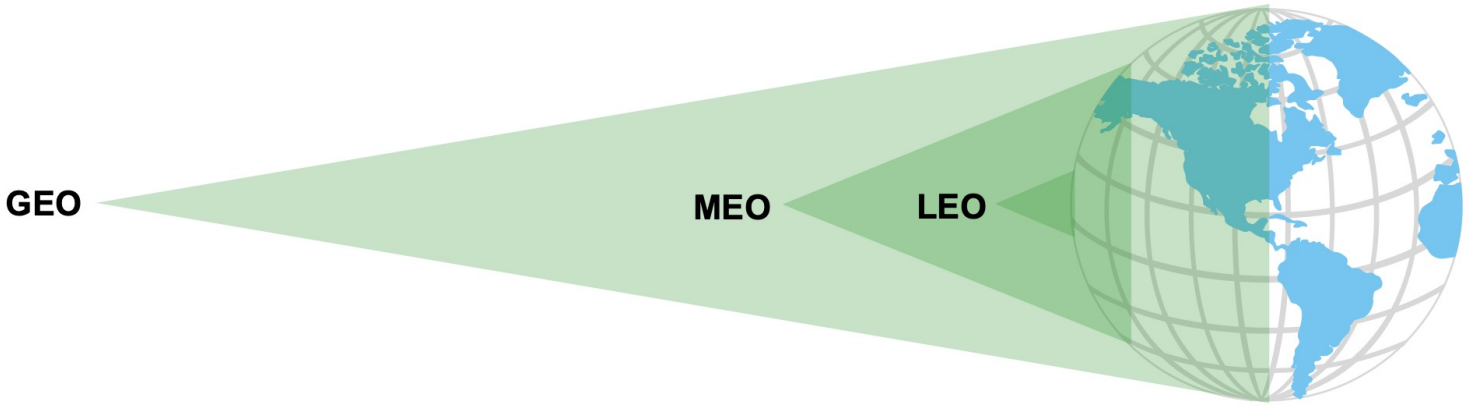
- **Height:** >100 km; global coverage
- **Latency:** High
- **Flexibility:** low flexibility; fixed orbits
- **Sensing:** RF and visual (depending on mission)
- **Typical use cases:** GNSS, IoT, PNT

# Satellite Systems



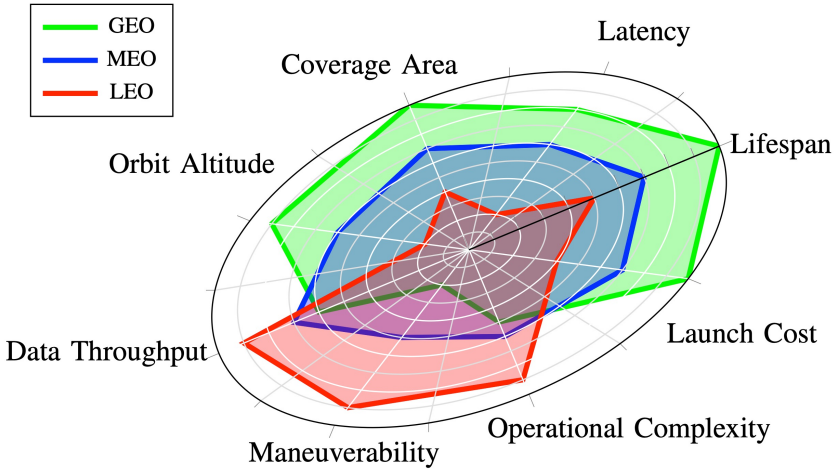
## Satellite

- **Height:** >100 km; global coverage
- **Latency:** High
- **Flexibility:** low flexibility; fixed orbits
- **Sensing:** RF and visual (depending on mission)
- **Typical Use Cases:** GNSS, IoT, PNT




| | Low Earth Orbit (LEO) | Medium Earth Orbit (MEO) | Geostationary Orbit (GEO) |
|---|---|---|---|
| **Altitude** | 160-1,500 km | 5,000-20,000 km | 35,786 km |
| **Latency** | Very low | Low | High |
| **Coverage** | Small | Large | Very large |

# Satellite Systems




## Localization Satellite Systems:

- GNSS
- LEO Satellites

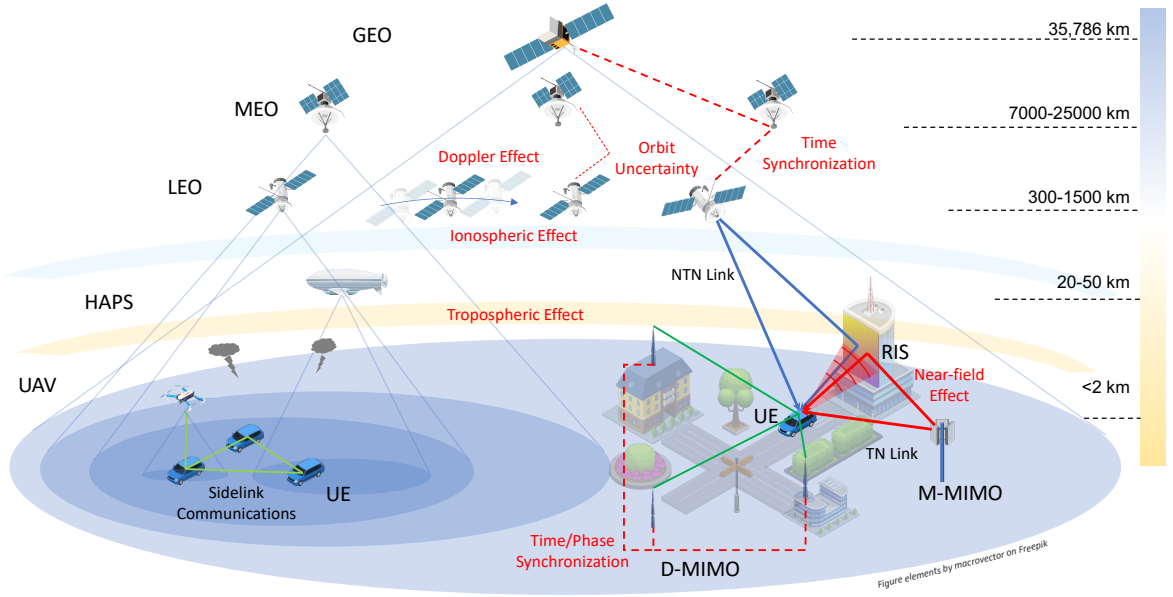
| Regime | Constellation | Orbital height | Quantity | Bands | Modulation | Services |
|---|---|---|---|---|---|---|
| GEO | Intelsat | 35,786 km | 52 | C, Ku, Ka | QPSK<br>8PSK<br>16APSK & 32APSK<br>QAM | Broadcasting<br>Data & communications<br>Mobile connectivity |
| | Eutelsat | 35,786 km | 36 | C, Ku, Ka | QPSK<br>8PSK<br>16APSK | Video & data services<br>Mobility<br>Government Services |
| | | | | | QPSK<br>8PSK<br>DVB-S2X | Broadcasting<br>Telecommunications<br>Broadband Internet |
| MEO | GPS | 20,180 km | 31 | L | BPSK, BOC | PNT |
| | GLONASS | 19,130 km | 24 | L | BPSK | PNT |
| | Galileo | 23,220 km | 30 | L | BOC, CBOC | PNT |
| | BeiDou | 21,530 km | 27<br>(+ 3 IGSO & 5 GEO) | L | QPSK, BOC | PNT |
| LEO | Xona | 525 km | 258 | L, C | - | PNT |
| | Starlink Gen.1<br>Starlink Gen.2 | 335.9 ∼ 570 km<br>340 ∼ 614 km | 11,926<br>30,000 | L, S, Ku, Ka, V | OFDM | Broadband Internet<br>& direct-to-cell services |
| | OneWeb | 1,200 km | 648 | Ku | OFDM | Broadband Internet |
| | Telesat Phase.1<br>Telesat Phase.2 | 1,015 ∼ 1,325 km | 298<br>1,373 | Ku, Ka | - | Broadband Internet |
| | Hongyan | 1,100 ∼ 1,175 km | 320 | Ka | - | Broadband Internet |
| | Kuiper | 590 ∼ 630 km | 3,236 | Ka | - | Broadband Internet |
| | Globalstar | 1,414 km | 48 | L, S | OQPSK | Voice and data services |
| | Orbcomm Gen.1<br>Orbcomm Gen.2 | 720 km<br>750 km | 36<br>18 | VHF | SDPSK | IoT & M2M communications |
| | Iridium NEXT | 781 km | 66 | L, Ka | QPSK | Voice and data<br>M2M communications |
| | Dove & SkySat (Planet) | 400 ∼ 600 km | 194 | Multispectral | - | Earth observation |
| | ICEYE | 560 ∼ 580 km | 27 | X | - | Earth observation |

Global Navigation Satellite System (GNSS)

Massive LEO Satellite Constellations

# Part III: NTN Characteristics and Localization Techniques

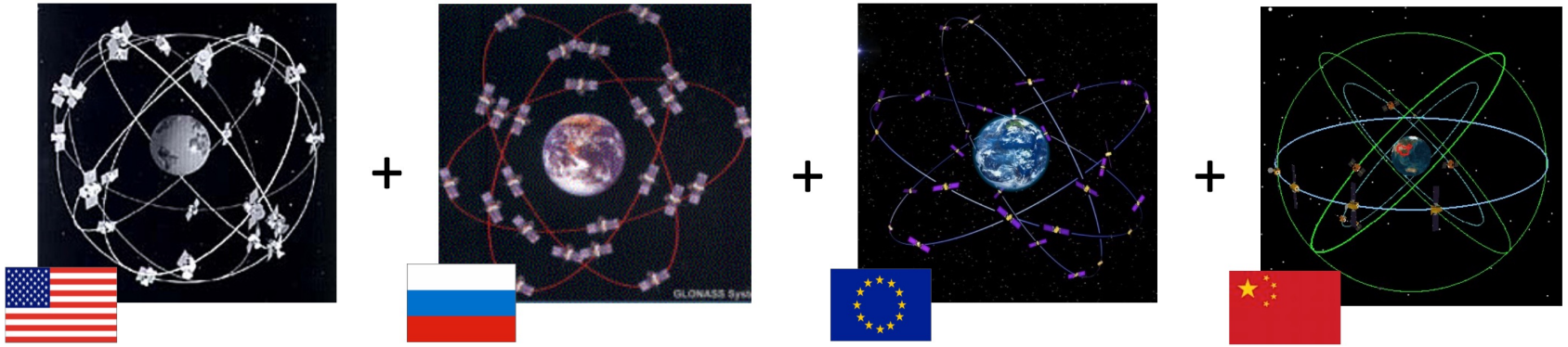
- Non-Terrestrial Systems
- Localization Based on GNSS
- Localization Based on LEO Satellites
- Case Studies



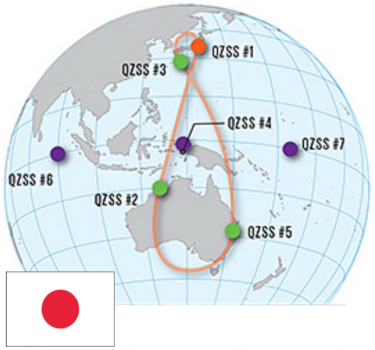
# GNSS Systems

**Global:**

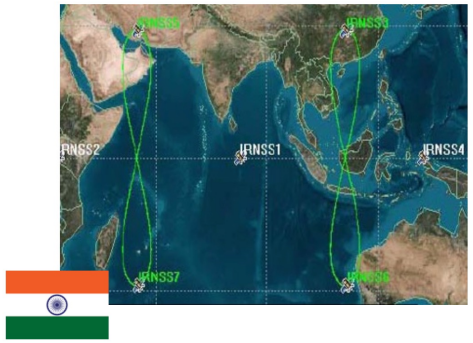
GPS + GLONASS + Galileo + BeiDou



**Regional:**




QZSS




NavIC

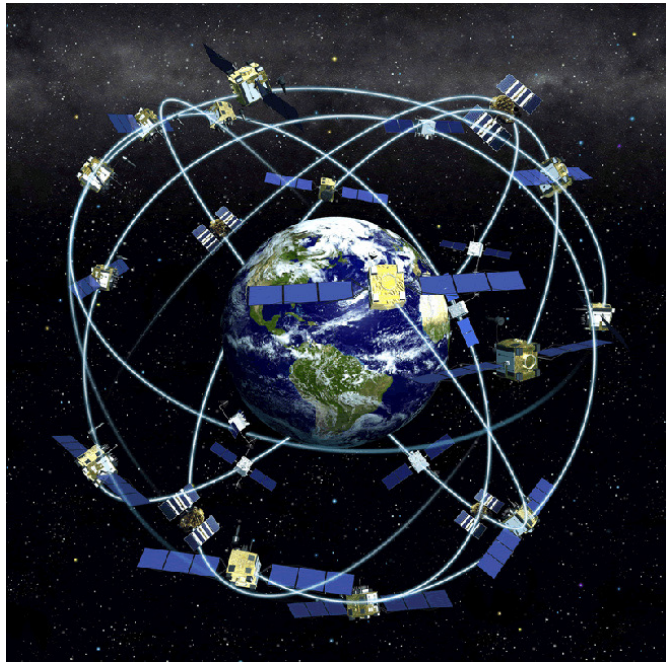
# Global GNSS Systems

| | GPS | GLONASS | Galileo | BeiDou |
|---|---|---|---|---|
| **Altitude** | 20,180 km | 19,130 km | 23,222 km | 21,150 km |
| **Period** | 11 h 58 min | 11 h 16 min | 14 h 5 min | 12 h 38 min |
| **Satellites** | 31 | 24 | 30 | 27 |
| **Frequencies** | 1.563--1.587 GHz (L1)<br>1.215--1.2396 GHz (L2)<br>1.164--1.189 GHz (L5) | 1.593--1.610 GHz (G1)<br>1.237--1.254 GHz (G2)<br>1.189--1.214 GHz (G3) | 1.559--1.592 GHz (E1)<br>1.164--1.215 GHz (E5a/b)<br>1.260--1.300 GHz (E6) | 1.561098 GHz (B1)<br>1.589742 GHz (B1-2)<br>1.20714 GHz (B2)<br>1.26852 GHz (B3) |
| **Signal** | CDMA | FDMA | CDMA | CDMA |

# GNSS Components



Control Segment



Space Segment



User Segment

# GNSS Components

## Control Segment

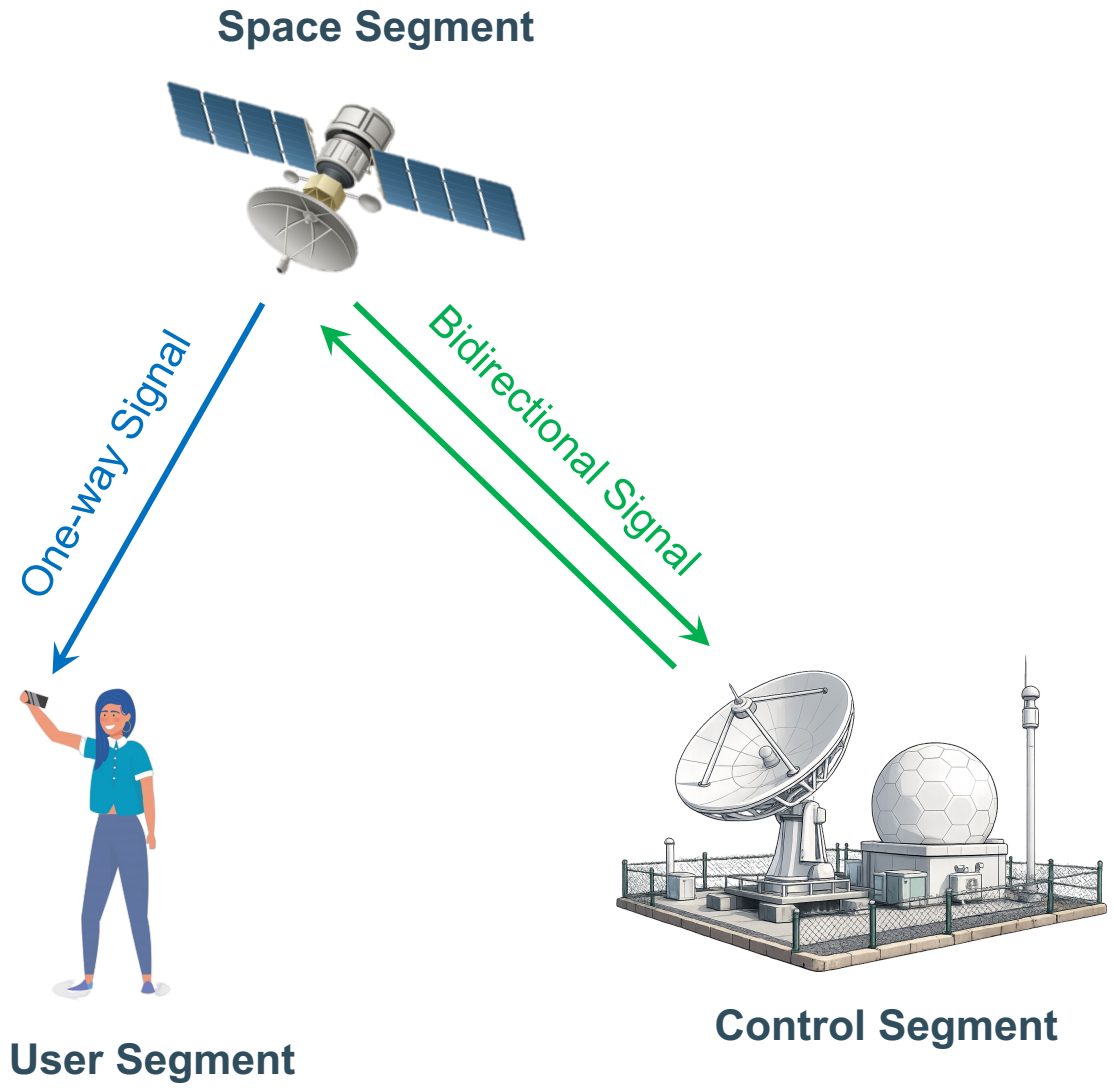
- Network of ground control stations worldwide
- Monitors satellite health and signal precision

## Space Segment

- Consist of the constellation of GNSS satellites
- Transmit signals to Earth

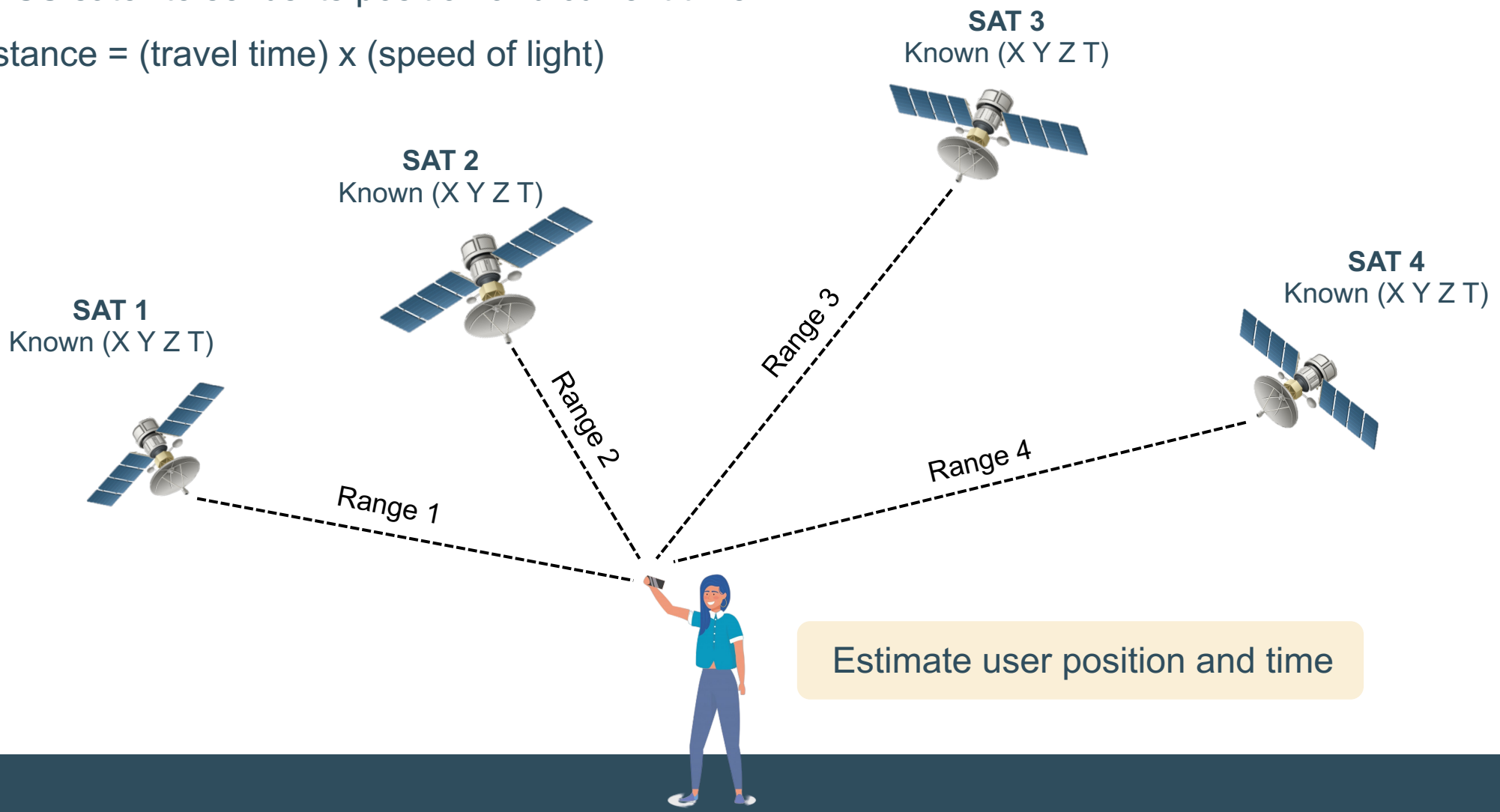
## User Segment

- Receivers in navigational units
- Process satellite signals to obtain PNT solutions



# How Does GNSS Work?

- Each GNSS satellite sends its position and current time
- User: distance = (travel time) x (speed of light)



# Typical GNSS Performance

| Positioning Technology | Accuracy (1-sigma) | Convergence Time | Coverage Area |
|---|---|---|---|
| **Single point positioning (SPP)** | < 10 m | Instantaneous | Global |
| **Differential GNSS (DGNSS) (Code-based)** | 1--5 m | Instantaneous | Local |
| **Real-time kinematic (RTK)** | < 1 dm | Instantaneous to a few minutes | Local |
| **Network RTK** | < 1 dm | < 10 minutes | Regional |
| **Precise point positioning (PPP)** | < 2 dm | A few minutes to hours | Global |
| **PPP-RTK** | < 1 dm | A few seconds to several minutes | Global |

# Example: Real-Time Kinematic (RTK) Localization

tropospheric delay

ionospheric delay

additive noise

geometrical range

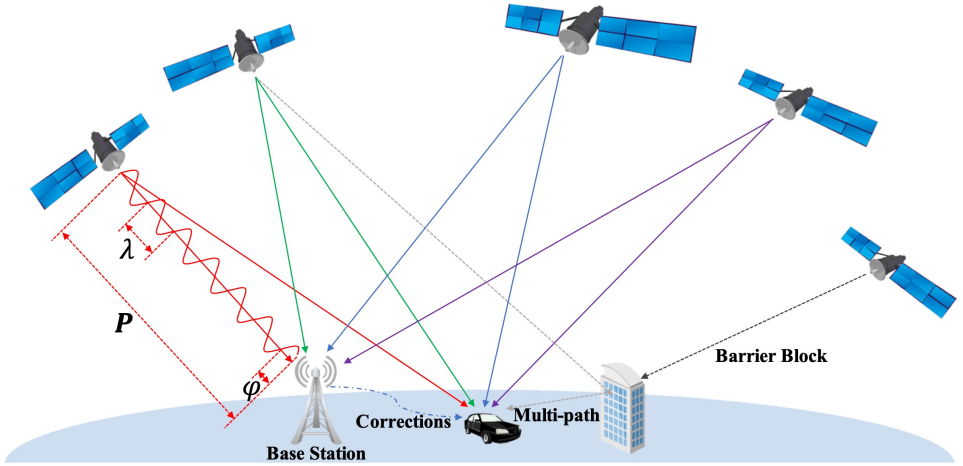
clock bias

**Pseudo range →** $P_u^n = \rho_u^n + I_u^n + T_u^n + c\,(\delta t_u - \delta t^n) + \varepsilon_u^n,$

**Carrier phase →** $\varphi_u^n = \rho_u^n + \lambda K_u^n - I_u^n + T_u^n + c\,(\delta t_u - \delta t^n) + \varsigma_u^n,$

GNSS carrier wavelength

carrier phase ambiguity (**integer**)




---

*1.* ***Single-difference (SD)*** *Between* ***BS*** *and* ***UE****:*

$$P_{ub}^n = P_u^n - P_b^n = \rho_{ub}^n + c\delta t_{ub} + \varepsilon_{ub}^n,$$
$$\varphi_{ub}^n = \varphi_u^n - \varphi_b^n = \rho_{ub}^n + \lambda K_{ub}^n + c\delta t_{ub} + \varsigma_{ub}^n.$$

Eliminate the satellite clock bias and atmospheric delays

[1] P. Teunissen and O. Montenbruck, Springer handbook of global navigation satellite systems. Springer, 2017.

# Example: Real-Time Kinematic (RTK) Localization

Note the far-field assumption : $\rho_{ub}^{n} = \rho_{u}^{n} - \rho_{b}^{n} = (\mathbf{h}_{u}^{n})^{\mathsf{T}}(\mathbf{p} - \mathbf{p}_{b})$

Satellite-UE LoS vector

*2. **Double-difference (DD)** Between Satellites:*

$$P_{ub}^{nm} = P_{ub}^{n} - P_{ub}^{m} = (\mathbf{h}_{u}^{n} - \mathbf{h}_{u}^{m})^{\mathsf{T}}(\mathbf{p} - \mathbf{p}_{b}) + \varepsilon_{ub}^{nm},$$
$$\varphi_{ub}^{nm} = \varphi_{ub}^{n} - \varphi_{ub}^{m} = (\mathbf{h}_{u}^{n} - \mathbf{h}_{u}^{m})^{\mathsf{T}}(\mathbf{p} - \mathbf{p}_{b}) + \lambda K_{ub}^{nm} + \varsigma_{ub}^{nm}.$$

Remove the receiver clock bias

$$\mathbf{H} = \left[(\mathbf{h}_{u}^{2} - \mathbf{h}_{u}^{1}), \ldots, (\mathbf{h}_{u}^{N} - \mathbf{h}_{u}^{1})\right]^{\mathsf{T}} \in \mathbb{R}^{(N-1)\times 3},$$
$$\mathbf{k} = \left[K_{ub}^{21}, K_{ub}^{31}, \ldots, K_{ub}^{N1}\right]^{\mathsf{T}} \in \mathbb{Z}^{N-1}.$$

**Integer constraint!**

*3. Concatenate all the **DD observations** as:*

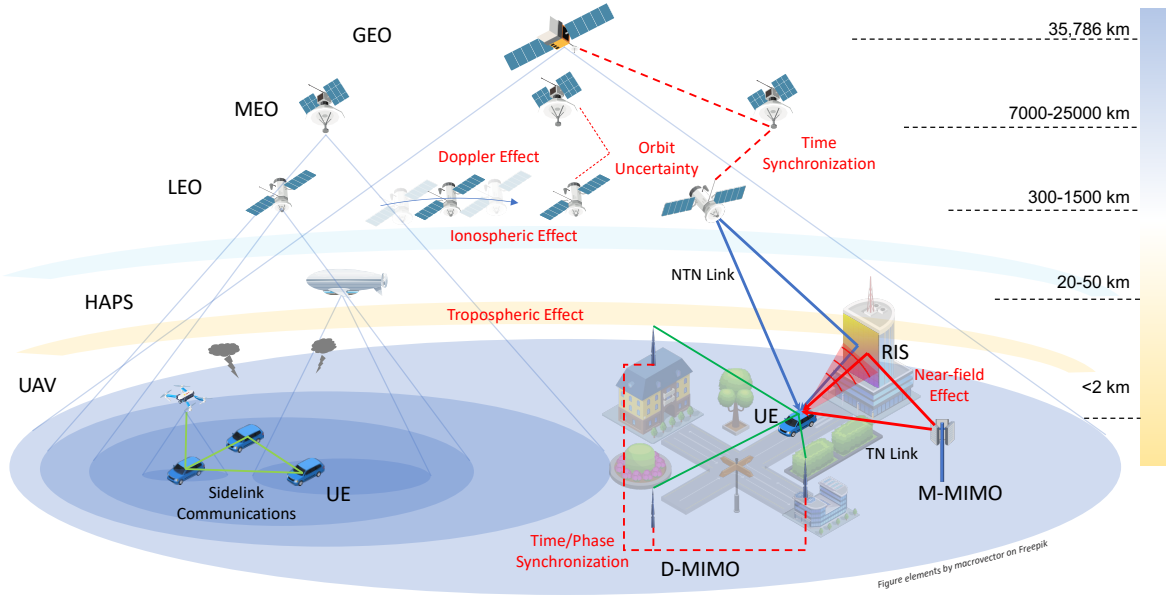
$$\mathbf{d} = [P_{ub}^{21}, P_{ub}^{31}, \ldots, P_{ub}^{N1}]^{\mathsf{T}} \in \mathbb{R}^{N-1},$$
$$\boldsymbol{\phi} = [\varphi_{ub}^{21}, \varphi_{ub}^{31}, \ldots, \varphi_{ub}^{N1}]^{\mathsf{T}} \in \mathbb{R}^{N-1}.$$

**We need to estimate**

$$\begin{bmatrix}\mathbf{d}\\ \boldsymbol{\phi}\end{bmatrix} = \underbrace{\begin{bmatrix}\mathbf{H}\\ \mathbf{H}\end{bmatrix}}_{\mathbf{G}}\mathbf{p} + \underbrace{\begin{bmatrix}\mathbf{0}\\ \lambda\mathbf{I}\end{bmatrix}}_{\mathbf{C}}\mathbf{k} + \underbrace{\begin{bmatrix}\mathbf{H}\\ \mathbf{H}\end{bmatrix}}_{\mathbf{b}}\mathbf{p}_{b} \in \mathbb{R}^{2N-2},$$

# Part III: NTN Characteristics and Localization Techniques

- Non-Terrestrial Systems
- Localization Based on GNSS
- Localization Based on LEO Satellites
- Case Studies

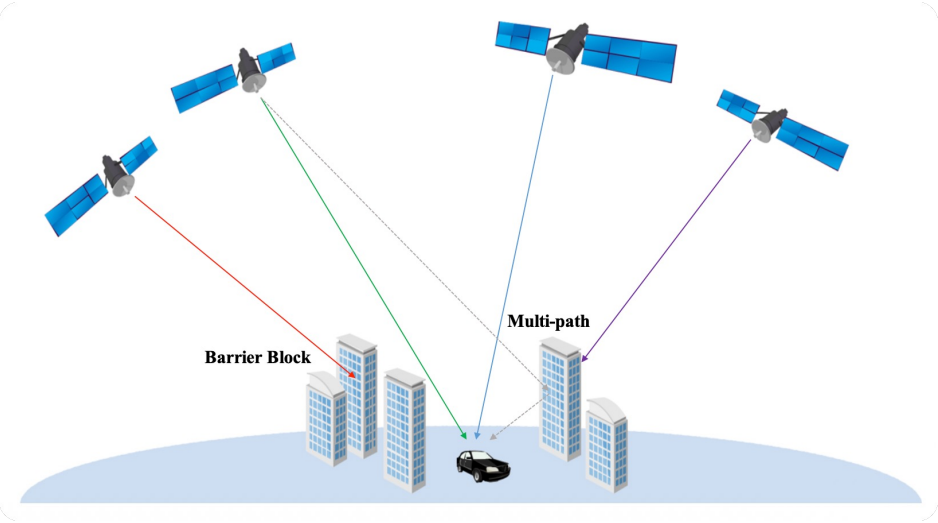


# Why Do We Need LEO Localization?

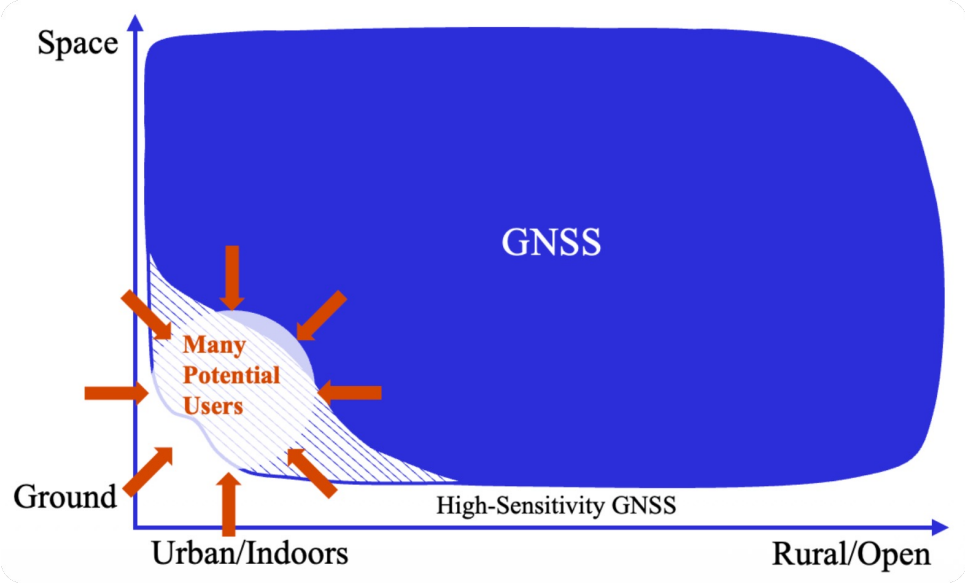
## Limitations of GNSS

- Accuracy declines in urban areas
- Long propagation distances → Weak received signals → Vulnerable to jamming and spoofing
- No return channel → No adaptive resource management

…




GNSS exhibits huge positioning errors **(up to 50 m)** in urban canyons




LEO satellites can complement GNSS by enhancing coverage, robustness, and positioning performance.

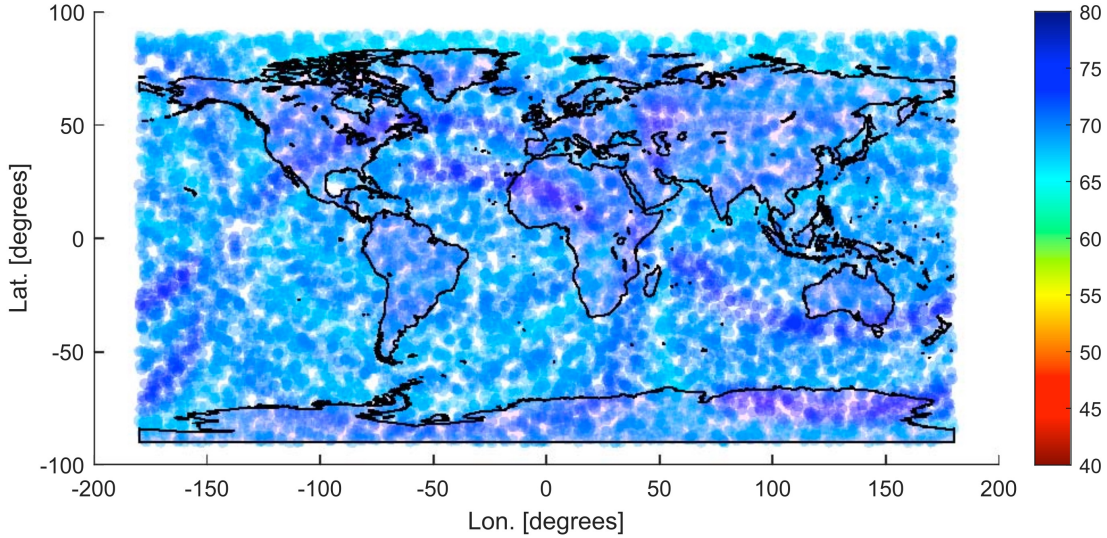
[1] M. M. Miller, A. Soloviev, M. Uijt de Haag, M. Veth, J. Raquet, T. J. Klausutis, and J. E. Touma, “Navigation in GPS denied environments: Feature-aided inertial systems,” AIR FORCE RESEARCH LAB EGLIN AFB FL MUNITIONS DIRECTORATE, Tech. Rep., 2010.
[2] X. Liu, T. Ballal, and T. Y. Al-Naffouri. "GNSS-based localization for autonomous vehicles: Prospects and challenges." Proc. 27th Eur. Signal Process. Conf. (EUSIPCO). 2019.

# LEO Satellites vs. GNSS

## Stronger Received Signal

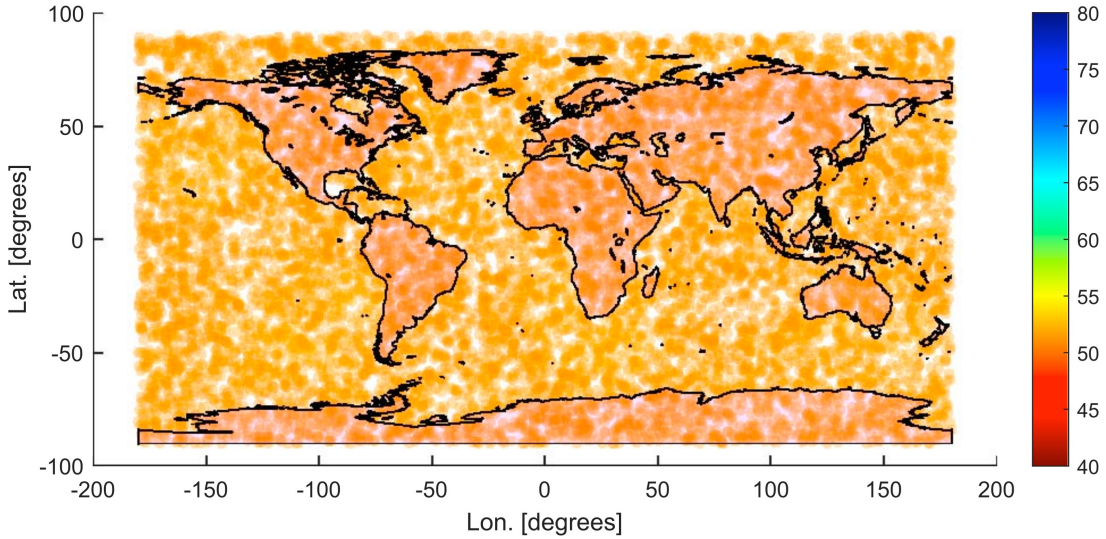
Carrier-to-noise density ratio **CN0** = $\frac{\textbf{Carrier power}}{\textbf{Noise power spectral density}}$

Average CN0 map for **Starlink** [1]




Mean CN0 = **69.9982 [dB-Hz]**

Average CN0 map for **Galileo** [1]




Mean CN0 = **51.2451 [dB-Hz]**

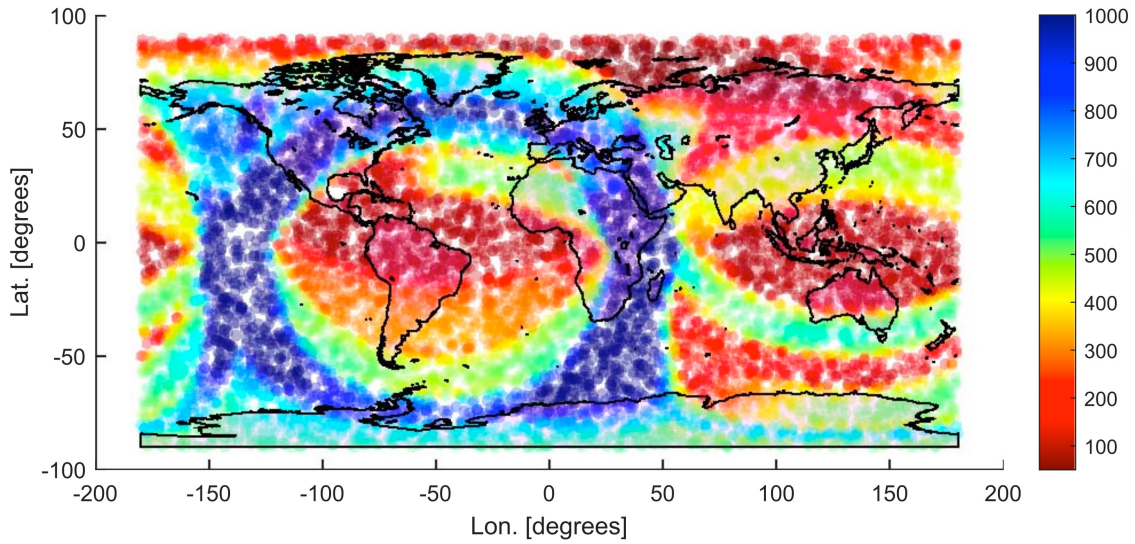
LEO signals are **10-20 dB-Hz** higher than GNSS signals!

[1] R. Morales Ferre, J. Praks, G. Seco-Granados and E. S. Lohan, "A Feasibility Study for Signal-in-Space Design for LEO-PNT Solutions With Miniaturized Satellites," in IEEE Journal on Miniaturization for Air and Space Systems, vol. 3, no. 4, pp. 171-183, Dec. 2022.

# LEO Satellites vs. GNSS

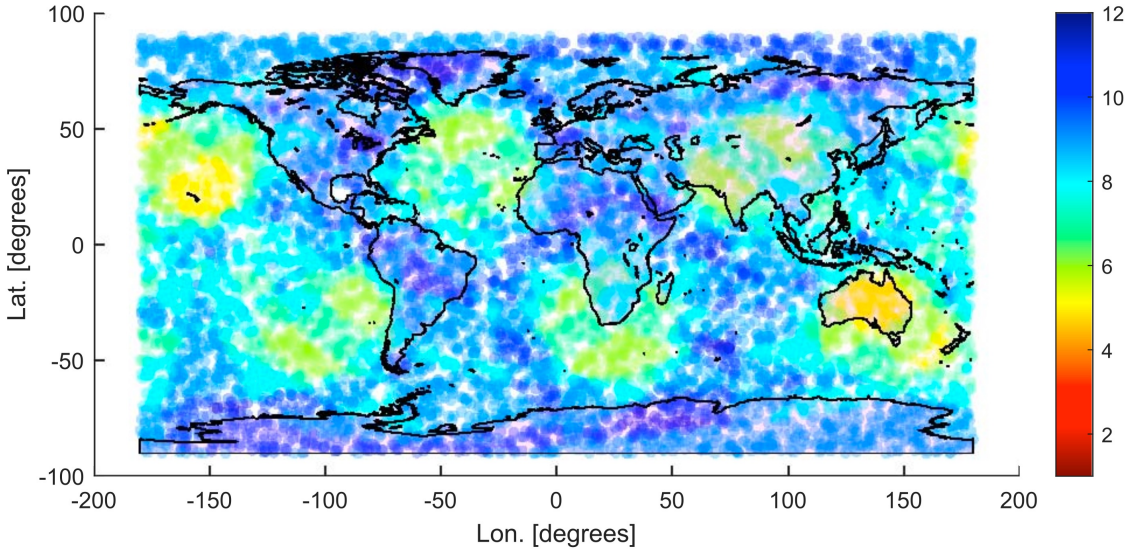
## Enhanced Visibility

Average satellite-in-view map for **Starlink** [1]




Mean satellite-in-view = **497.5197**

Average satellite-in-view map for **Galileo** [1]

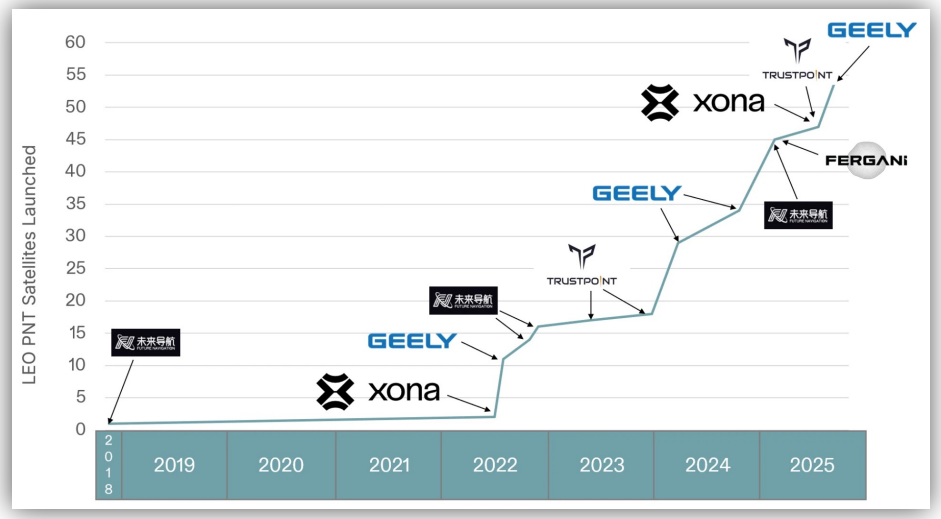



Mean satellite-in-view = **8.3479**

Much more satellites are visible!

# Recent Progress Towards LEO-PNT

- In Europe, **European Space Agency (ESA)** has advanced LEO-PNT mainly through the **Celeste program** [1], [2].
- In USA, **Xona** launched a validation satellite for LEO-PNT system in June 2025 [3]. **SpaceX** has stated that **Starlink** terminals can provide meter-level positioning and nanosecond-level timing [4].





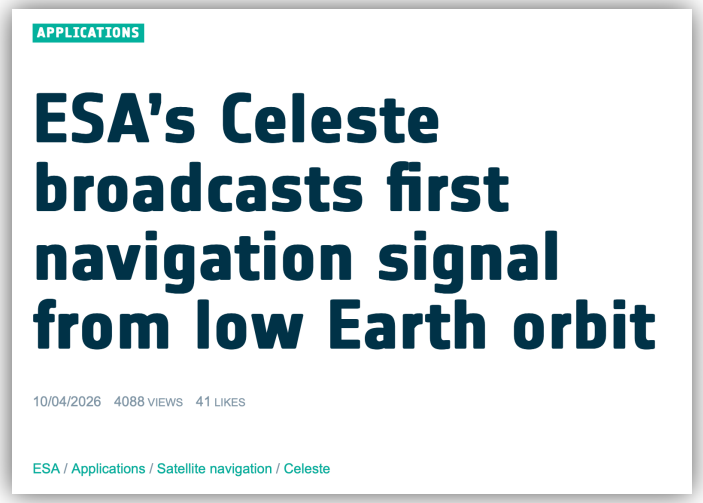
[1] European Space Agency: Introducing Celeste. Accessed: 2026-05-11 (2026). https://www.esa.int/Applications/Satellite_navigation/Celeste/Introducing_Celeste

[2] European Space Agency: ESA's Celeste Broadcasts First Navigation Signal from Low Earth Orbit. Accessed: 2026-05-11(2026). https://www.esa.int/Applications/Satellite_navigation/Celeste/ESA_s_Celeste_broadcasts_first_navigation_signal_from_low_Earth_orbit

[3] Xona Space Systems: Launching Pulsar-0, Our First Production-Class Satellite. Accessed: 2026-05-11 (2025). https://www.xonaspace.com/news/launching-pulsar-0

[4] GPS World: SpaceX Shares Starlink's Role in Strengthening U.S. PNT Resilience in Response to FCC Filing. Accessed: 2026-05-11 (2025). https://www.gpsworld.com/spacex-shares-starlinks-role-in-strengthening-us-pnt-resilience-in-response-to-fcc-filing/

[5] GPS World: The rise of LEO PNT. Accessed: 2026-05-11 (2026). https://www.gpsworld.com/the-rise-of-leo-pnt/

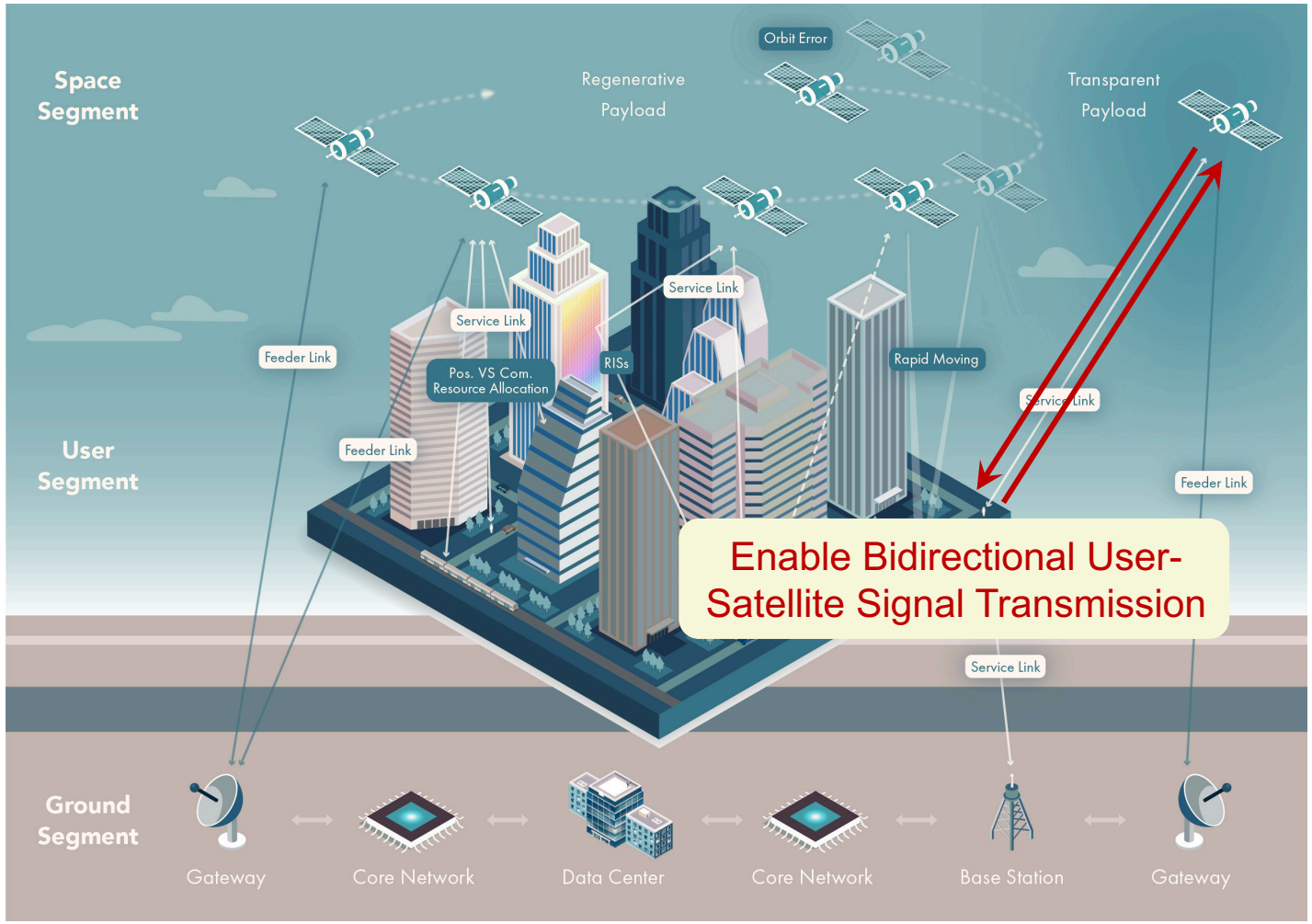


APPLICATIONS

**ESA's Celeste broadcasts first navigation signal from low Earth orbit**

10/04/2026 4088 VIEWS 41 LIKES

ESA / Applications / Satellite navigation / Celeste

# How Does LEO Satellites Work?




- Space segment, user segment, and ground segment.
- Service links and feeder links.
- Transparent payload and regenerative payload.
- Enable bidirectional signals.

[1] J. Ma, P. Zheng, X. Liu, Y. Zhang, and T. Y. Al-Naffouri, “Integrated Positioning and Communication via LEO Satellites: A Contemporary Overview,” npj Wireless Technology, vol. 2, no. 1, Art. no. 76, 2026.

# LEO Systems

## Very heterogeneous!

### D2D/NTN Systems

- **Target:** low-rate mobile services for handheld devices and low-complexity user terminals.
- **Spectrum:** mobile satellite service (MSS) spectrum, e.g., L/S band or other sub-3 GHz bands.
- **Traditional NTN Systems:** Iridium and Globalstar, with dedicated MSS spectrum.
- **Recent NTN Systems:** AST SpaceMobile, Starlink direct-to-cell (D2C), with direct cellular air interface compatibility.

### Broadband Systems

- **Target:** high-throughput connectivity.
- **Spectrum:** Ku/Ka band.
- **Satellite Architecture:** phased-array antennas, multibeam transmission, high-capacity feeder links, and inter-satellite links.
- **User Terminal Architecture:** more capable than handheld devices, with directional or electronically steered apertures.
- **Representative Systems:** Starlink Gen1/Gen2, OneWeb, and Kuiper.

### Dedicated PNT Systems

- **Target:** positioning, navigation, and timing services.
- **Spectrum:** L/S band.
- **Waveform:** GNSS-compatible dedicated PNT waveform.
- **User Terminal Architecture:** compatible with GNSS receivers.
- **Type 1:** native LEO-PNT systems, such as Xona Pulsar and Celeste.
- **Type 2:** enhanced PNT services built on existing systems such as Iridium PNT.

# LEO Systems

## No universal signal model!

**Table 1** Representative LEO Satellite Systems

| Type | Name | Orbit | Count | Bands | Waveform / Air interface | PNT / COM |
|---|---|---|---|---|---|---|
| D2D | AST SpaceMobile | 515 / 740 km | – | low-/mid-band | 4G/5G D2D | COM |
| | Starlink D2C | 550 km | 400 | sub-3 GHz | LTE-based D2C | COM |
| | Lynk | 479–532 km | – | cellular spectrum | GSM/LTE/5G phone-compatible | COM |
| | Iridium | 780 km | 66 | L | TDMA/FDMA-type MSS air interface | COM (+PNT layer) |
| | Globalstar | 1414 km | 48 | L/S | CDMA | COM |
| Broadband | Starlink Gen. 1 | 550 km | 11,926 | Ku/Ka/E | OFDM | COM |
| | Starlink Gen. 2 | 340–614 km | 29,988 | Ku/Ka | OFDM | COM |
| | OneWeb | 1200 km | 648 | Ku/Ka | – | COM |
| | Kuiper | 590–630 km | 3000 | Ka | – | COM |
| Dedicated / Enhanced PNT | Xona Pulsar | 1080 km | 258 | dual L (X1/X5) | GNSS-compatible dedicated PNT waveform | PNT |
| | ESA Celeste | 500–600 km | 11 | L/S | Dedicated dual-frequency navigation signal | PNT |

Communication signals

GNSS signals

# Challenges and Opportunities in LEO Localization

**Challenges:**

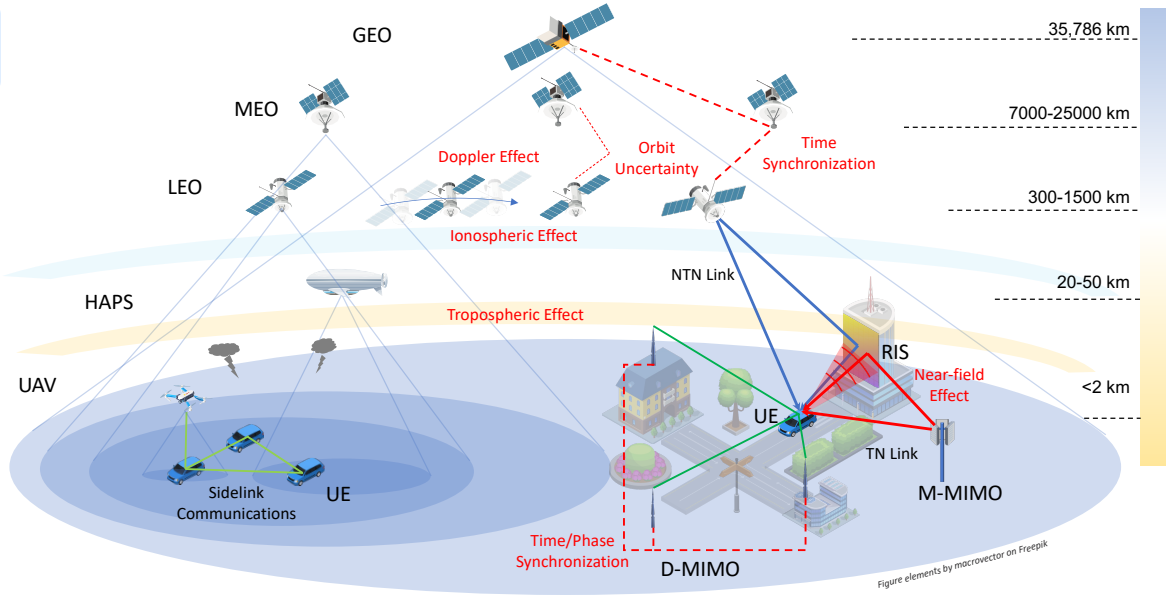
- Orbital Errors
- Frequent Handover
- Heterogeneous Signal Processing

**Opportunities:**

- Stronger Doppler
- Integrated Positioning and Communication (IPAC)
- Integrated TN and NTN Localization

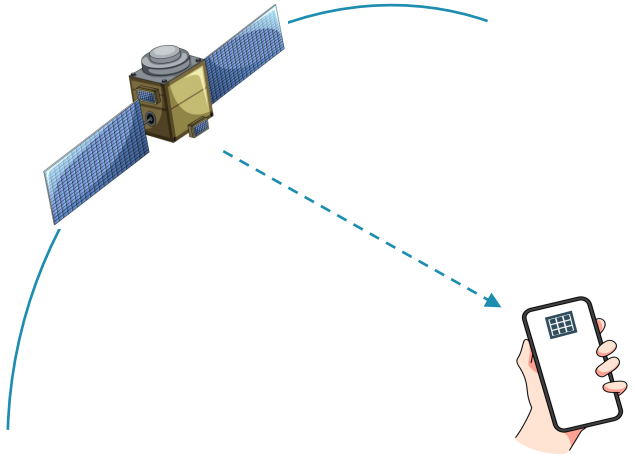
# Part III: NTN Characteristics and Localization Techniques

- Non-Terrestrial Systems
- Localization Based on GNSS
- Localization Based on LEO Satellites
- Case Studies



# Case Study 1: LEO Positioning Under Orbital Errors

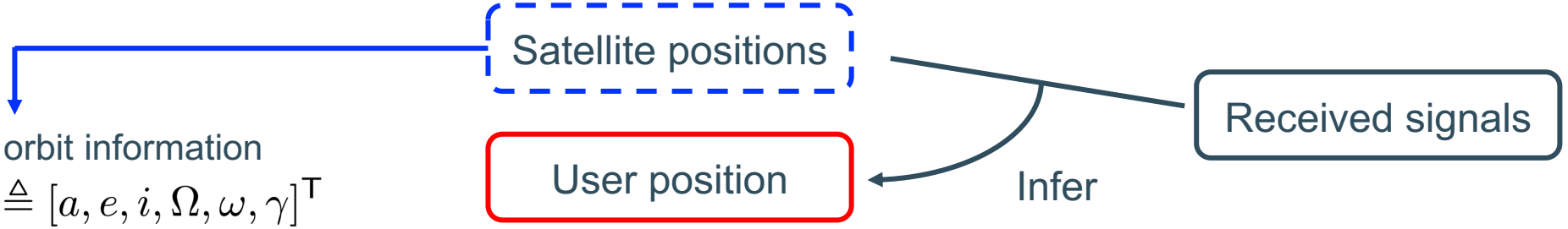
## How Do Orbital Errors Impact Localization Performance?



- **Signal Reception:**




- **Localization:**




  - Obtained via orbit information

$$\mathbf{o}_{\text{sat}} \triangleq [a, e, i, \Omega, \omega, \gamma]^{\mathsf{T}}$$

  - What if they are **inaccurate**?

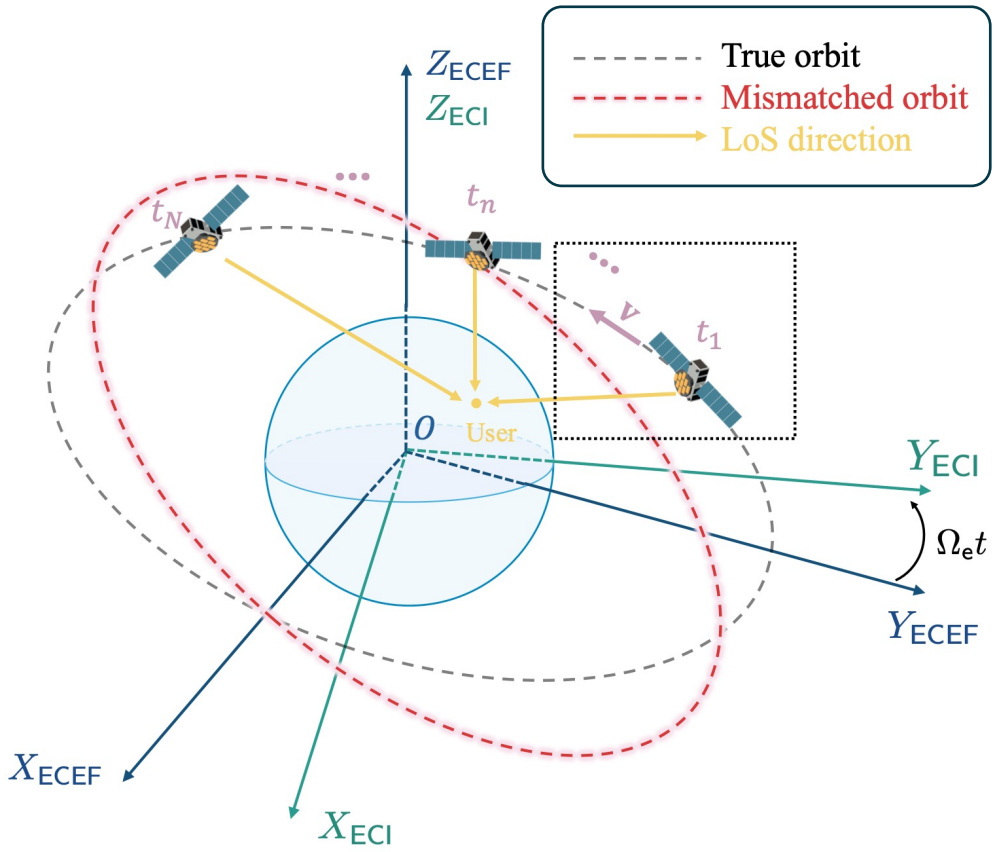
# Case Study 1: LEO Positioning Under Orbital Errors

**Why orbital errors are more severe for LEO satellites?**

- Low altitude → high orbital speed → rapid position changes
- Stronger perturbations (atmospheric drag, gravity irregularities, solar effects, etc)

**Motivation**

- ✓ LEO satellites serve as location references for localizing users
- ✓ Inaccurate satellite position information can degrade user localization performance



# Case Study 1: LEO Positioning Under Orbital Errors

Now, we show how to determine the satellite's position in the ECI frame at time $t$ given $\mathbf{o}_{\text{sat}}$. To simplify the notation, we use $\Omega_t$, $\omega_t$ and $\gamma_t$ to denote time-dependent orbital elements. Given the Earth's non-sphericity factor $J_2$, the Earth's radius $R_e$, the Earth's mass $M_e$, gravitational constant $\mu$ and the initial values of $(\Omega_0, \omega_0)$ at time $t_0$, we first compute the perturbation coefficient $\kappa$ and the mean motion correction $\bar{m}$ as

$$\kappa = \frac{1.5 \cdot J_2 R_e^2}{a^2(1-e^2)^2}, \tag{8}$$

$$\bar{m} = \sqrt{\frac{\mu M_e}{a^3}} \left(1 + \kappa(1 - 1.5 \cdot \sin^2 i)\sqrt{1-e^2}\right). \tag{9}$$

The RAAN and argument of periapsis at $t$ are then updated as

$$\Omega_t = \Omega_0 - t \cdot \kappa \cdot \bar{m} \cdot \cos i, \tag{10}$$

$$\omega_t = \omega_0 + t \cdot \kappa \cdot \bar{m}(2 - 2.5 \cdot \sin^2 i). \tag{11}$$

Next, the mean anomaly $\bar{M}$ is computed and then used to solve Kepler's equation given by the following (12) to obtain the eccentric anomaly $E_t$:

$$\bar{M} = \bar{M}_0 + \bar{m}(t - t_0), \quad E_t - e \sin E_t = \bar{M}. \tag{12}$$

The true anomaly at time $t$ can be obtained by solving the following equation:

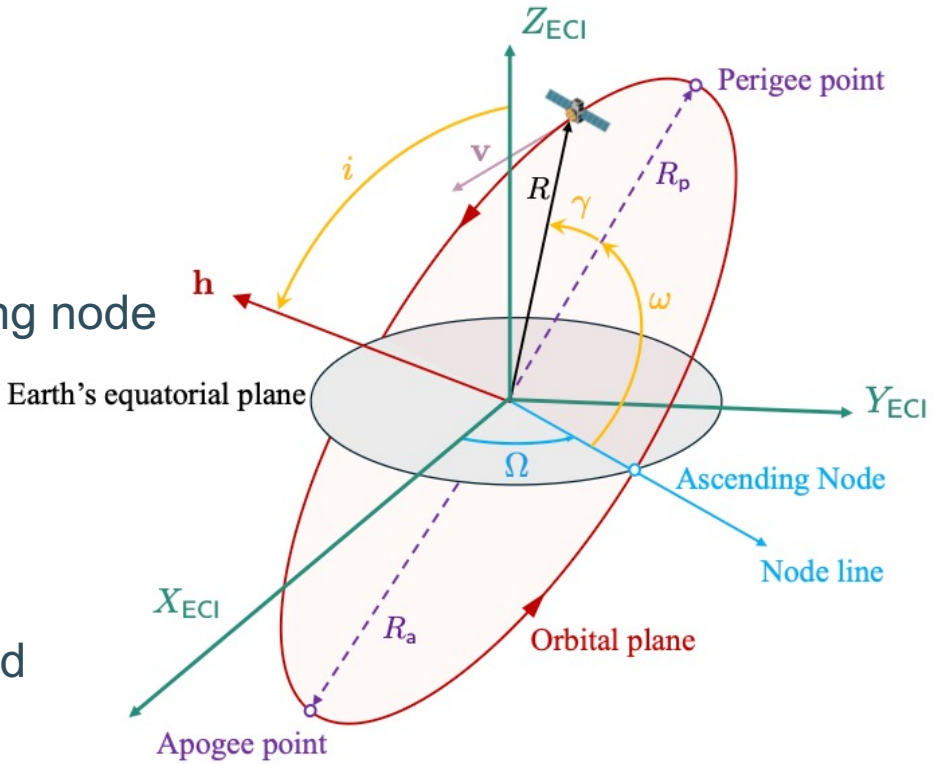
$$\tan(\gamma_t/2) = \sqrt{(1+e)/(1-e)} \tan(E_t/2). \tag{13}$$

Finally, the satellite's distance from the Earth's center and its position in ECI coordinates are calculated as:

$$R_t = \frac{a(1-e^2)}{1 + e\cos(\gamma_t)}, \tag{14}$$

$$\mathbf{p}_{\text{ECI},t} = R_t \begin{bmatrix} \cos(\omega_t+\gamma_t)\cos\Omega_t - \sin(\omega_t+\gamma_t)\sin\Omega_t \cos i \\ \cos(\omega_t+\gamma_t)\sin\Omega_t + \sin(\omega_t+\gamma_t)\cos\Omega_t \cos i \\ \sin(\omega_t+\gamma_t)\sin i \end{bmatrix}. \tag{15}$$

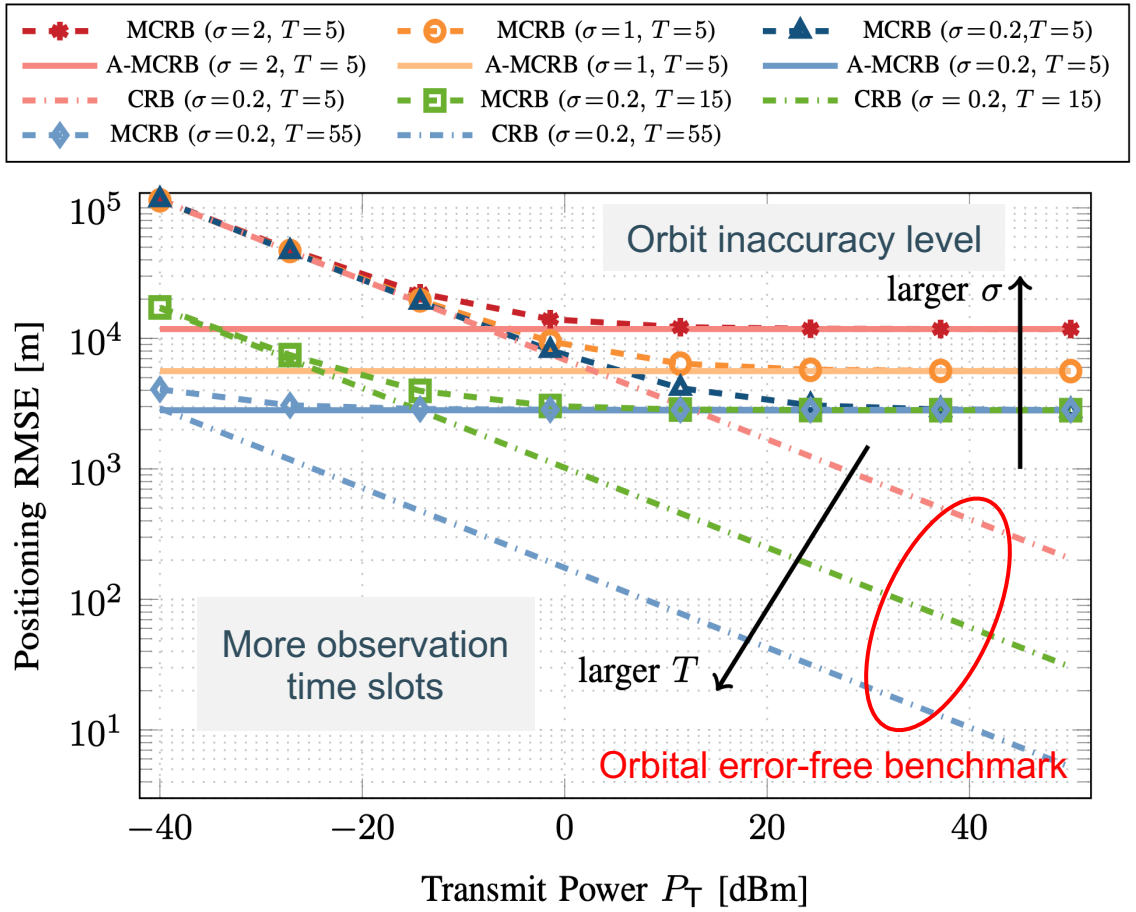
- Six Keplerian elements fully determine a satellite orbit [1]: $\mathbf{o}_{\text{sat}} \triangleq [a, e, i, \Omega, \omega, \gamma]^{\mathsf{T}}$
  - Semi-major axis
  - Eccentricity
  - Inclination
  - Right ascension of the ascending node
  - Argument of periapsis
  - True anomaly
- Obtaining LEO satellite positions usually relies on infrequently updated two-line-element (TLE) data [2].





[1] J. Ma, P. Zheng, X. Liu, Y. Zhang, A. A. Nasir, and T. Y. Al-Naffouri, "Orbital Errors in LEO Satellite Positioning: Modeling, Analysis, and Bayesian Calibration," arXiv preprint arXiv:2511.06060, 2026.
[2] K. Selvan, A. Siemuri, F. S. Prol, P. V¨ alisuo, M. Z. H. Bhuiyan, and H. Kuusniemi, "Precise orbit determination of LEO satellites: a systematic review," GPS Solutions, vol. 27, no. 4, p. 178, 2023.

# Case Study 1: Theoretical Insights

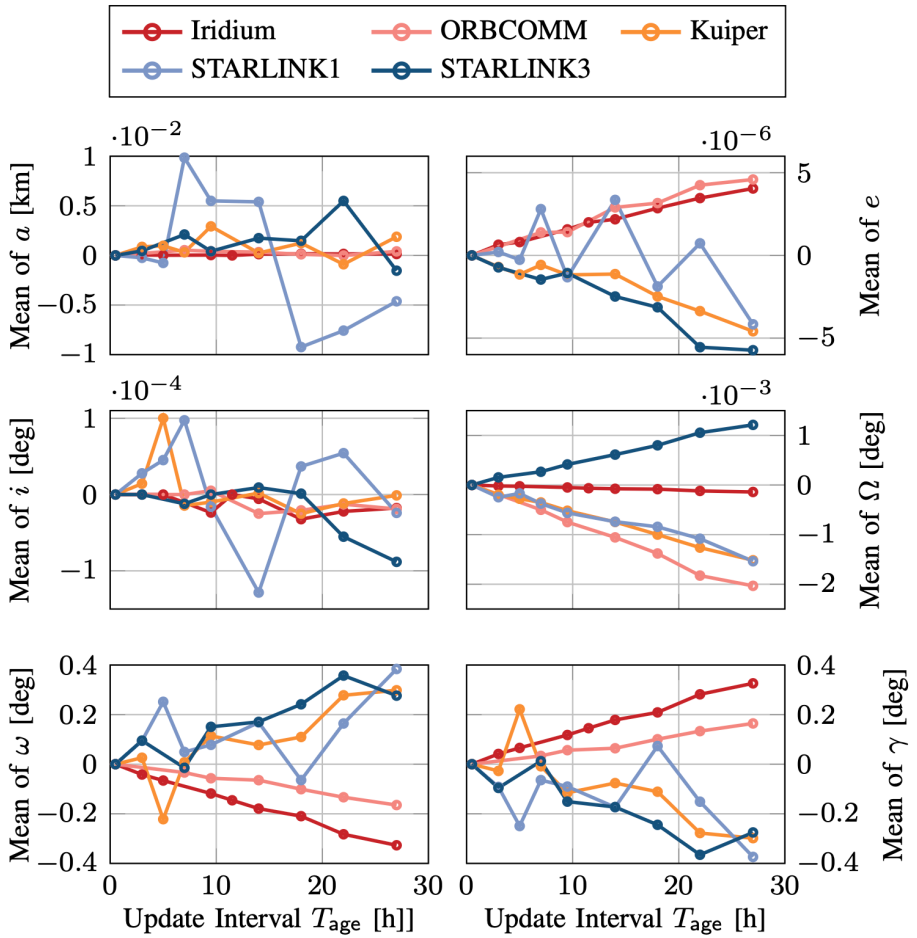



**How Do Orbital Errors Impact Localization Performance?**

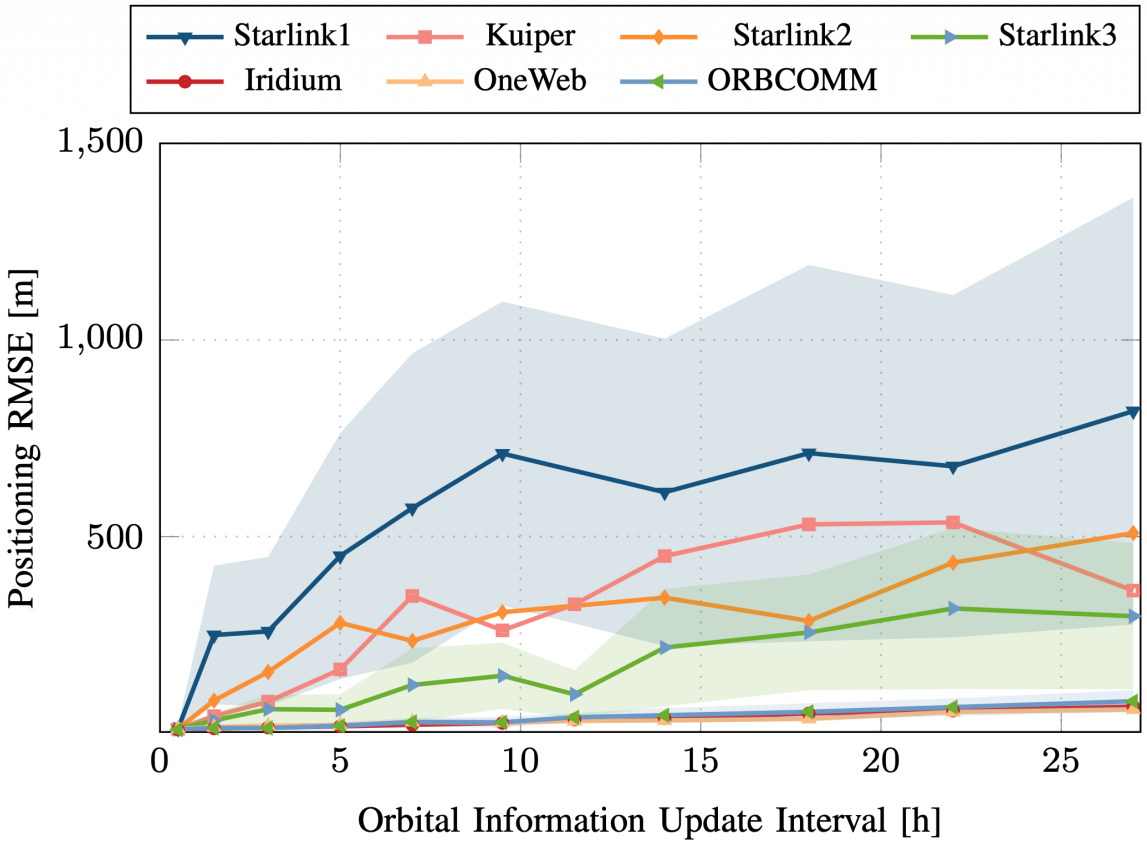
- ✓ Orbital-error-induced model mismatch limits the achievable localization accuracy.
- ✓ This impact cannot be mitigated by increasing power and observations.

# Case Study 1: Real Data Evaluation

## Error accumulation in the six orbital elements




## Error accumulation in the positioning result



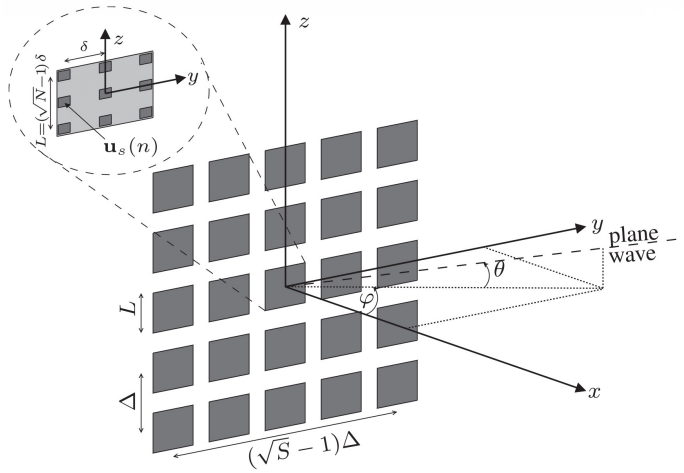

**LEO orbital calibration is necessary!**

# Case Study 2: Integrated Positioning and Communication (IPAC)

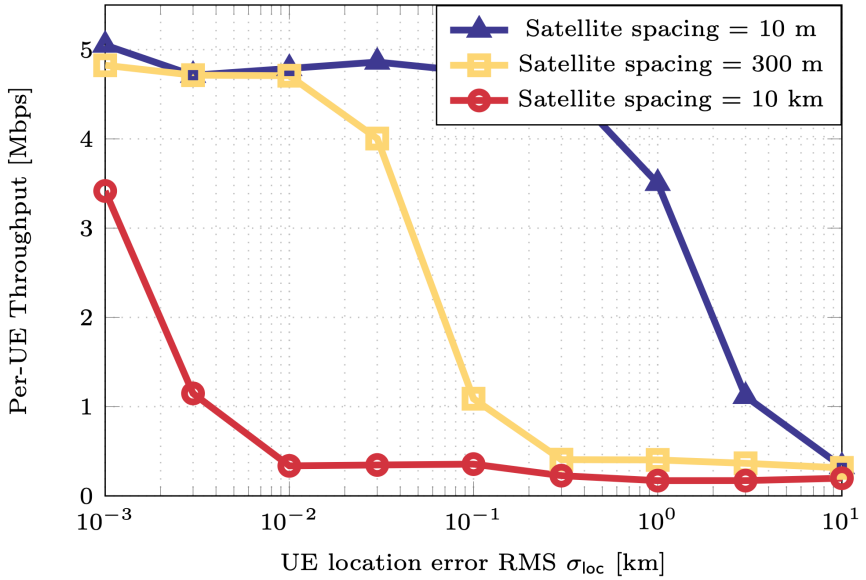
- ❑ LEO satellites possess both positioning and communication capabilities.
- ❑ How can these two functionalities mutually benefit each other?

## Example A: Location Information Helps Communication

- ➢ Rapidly varying channels → Challenging to estimate channel state information (CSI) → **How to do beamforming?**
- ➢ Satellite channels are LoS-dominated → **UE location** + **Satellite ephemeris**: Determine LoS channel predictively




Formation of Arrays (FoA) with a Cluster of Mobile Satellites [1]




- ✓ Having rough UE location information can be sufficient to do beamforming, as the footprint of satellite beam is large.
- ✓ The positioning accuracy requirement also depends on satellite formation.


[1] R. De Gaudenzi, G. Bacci, M. Luise, L. Sanguinetti and P. Angeletti, "Applicability of CF-MIMO Precoding to a Formation of Arrays (FoA) for Mobile Satellite Communications," in IEEE Transactions on Aerospace and Electronic Systems, vol. 61, no. 5, pp. 11069-11087, Oct. 2025.

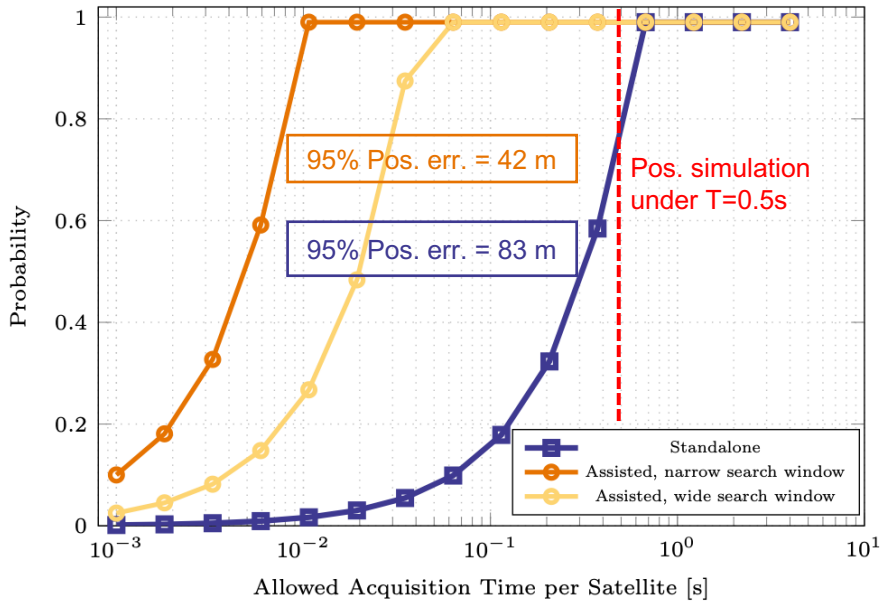
[2] [1] J. Ma, P. Zheng, X. Liu, Y. Zhang, and T. Y. Al-Naffouri, "Integrated Positioning and Communication via LEO Satellites: A Contemporary Overview," npj Wireless Technology, vol. 2, no. 1, Art. no. 76, 2026.

# Case Study 2: Integrated Positioning and Communication (IPAC)

- ❑ LEO satellites possess both positioning and communication capabilities.
- ❑ How can these two functionalities mutually benefit each other?

## Example B: Communication Signal Helps Positioning

- ➢ Large delay/Doppler/code uncertainty → Difficult LEO-PNT signal acquisition
- ➢ Communication information → **Coarse delay/Doppler prior**→ higher acquisition probability and lower positioning error




- ✓ The simulated accuracy reflects a cold-start, acquisition-limited scenario based on range observation.
- ✓ Having communication-side prior information can reduce the LEO-PNT acquisition search space, improving acquisition probability under limited acquisition time.
- ✓ Higher acquisition probability can provide more valid satellites, leading to lower positioning error.

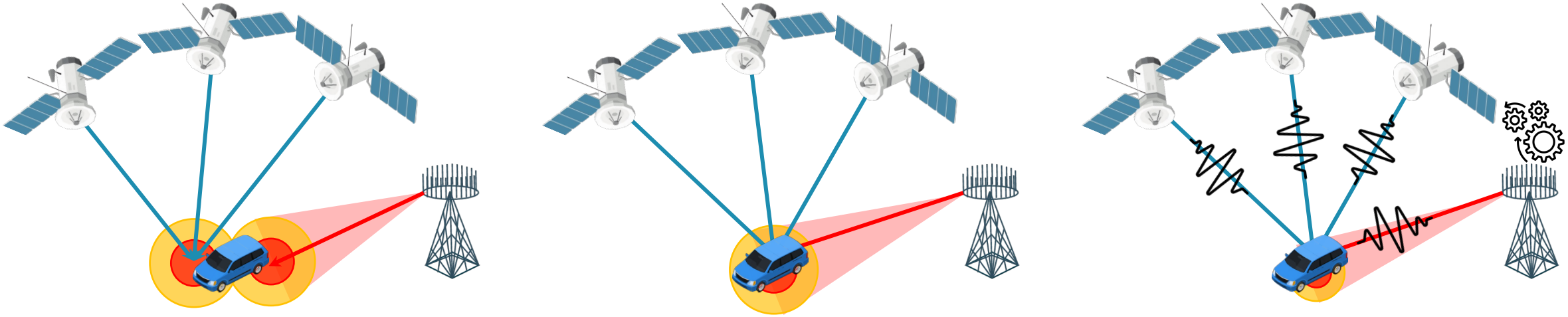
[1] [1] J. Ma, P. Zheng, X. Liu, Y. Zhang, and T. Y. Al-Naffouri, "Integrated Positioning and Communication via LEO Satellites: A Contemporary Overview," npj Wireless Technology, vol. 2, no. 1, Art. no. 76, 2026.

# Part IV: TN-NTN Integrated Localization

Dr. Pinjun Zheng

CHALMERS
UNIVERSITY OF TECHNOLOGY
UBC

# NTN-TN Integration levels



Position estimates — Loose Integration — Parameter estimates — Tight Integration — Radio signals — Ultra-tight Integration

Complexity and Accuracy

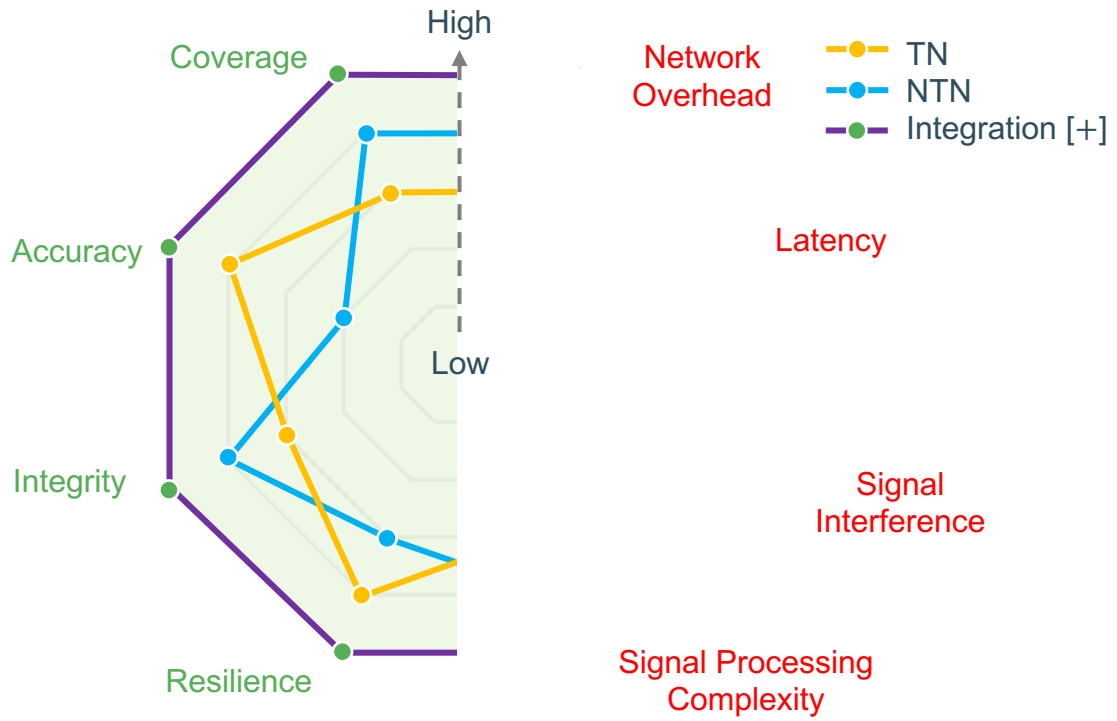
# NTN-TN Integration Opportunities and Challenges

**Opportunities:**

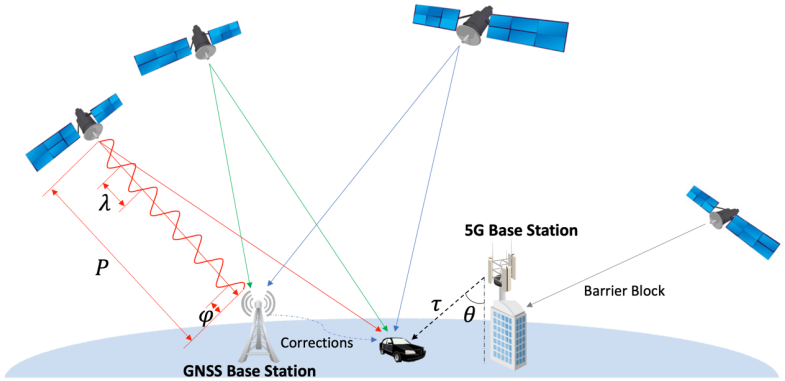
- Coverage
- Accuracy
- Integrity
- Resilience

**Challenges:**

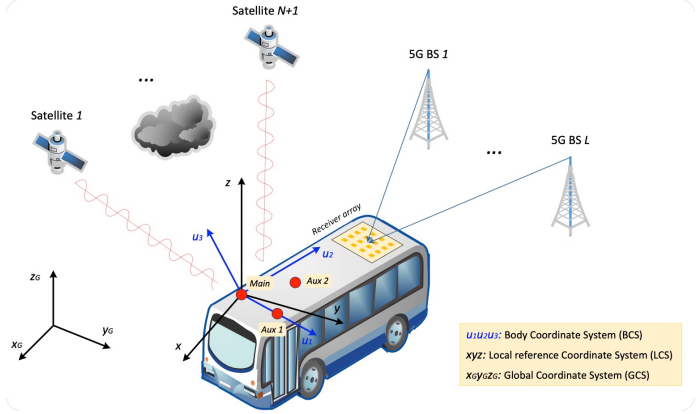
- Network overhead
- Latency
- Signal interference
- Signal processing complexity



# New Scenarios Underexplored

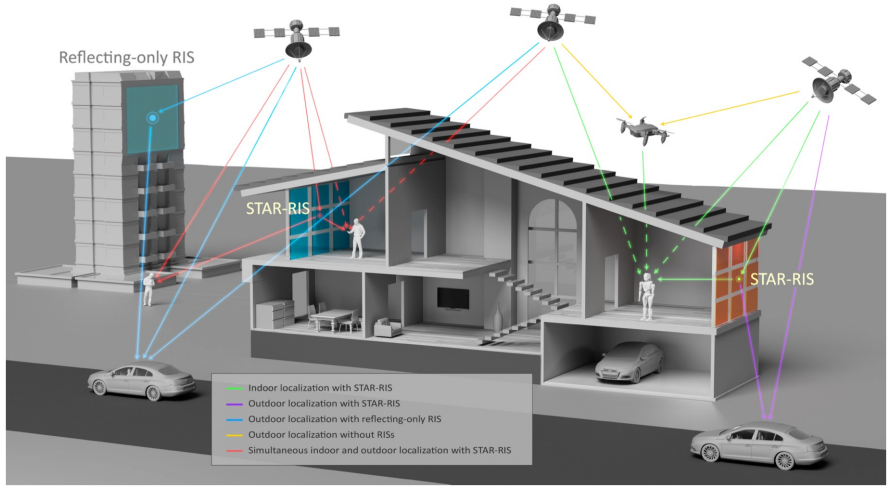



5G BS-aided GNSS Positioning [1]




5G BS-aided GNSS Attitude Determination [2]

...




Reconfigurable intelligent surface-aided localization and tracking [3], [4]


[1] P. Zheng, X. Liu, T. Ballal, and T. Y. Al-Naffouri, “5G-aided RTK positioning in GNSS-deprived environments,” EUSIPCO, 2023.

[2] P. Zheng, X. Liu, T. Ballal, and T. Y. Al-Naffouri, “Attitude Determination in Urban Canyons: A Synergy between GNSS and 5G Observations,” ION GNSS+ 2023.

[3] P. Zheng, X. Liu, and T. Y. Al-Naffouri, “LEO- and RIS-Empowered User Tracking: A Riemannian Manifold Approach,” in IEEE Journal of Selected Areas in Communications, vol. 42, no. 12, pp. 3445-3461, 2024.

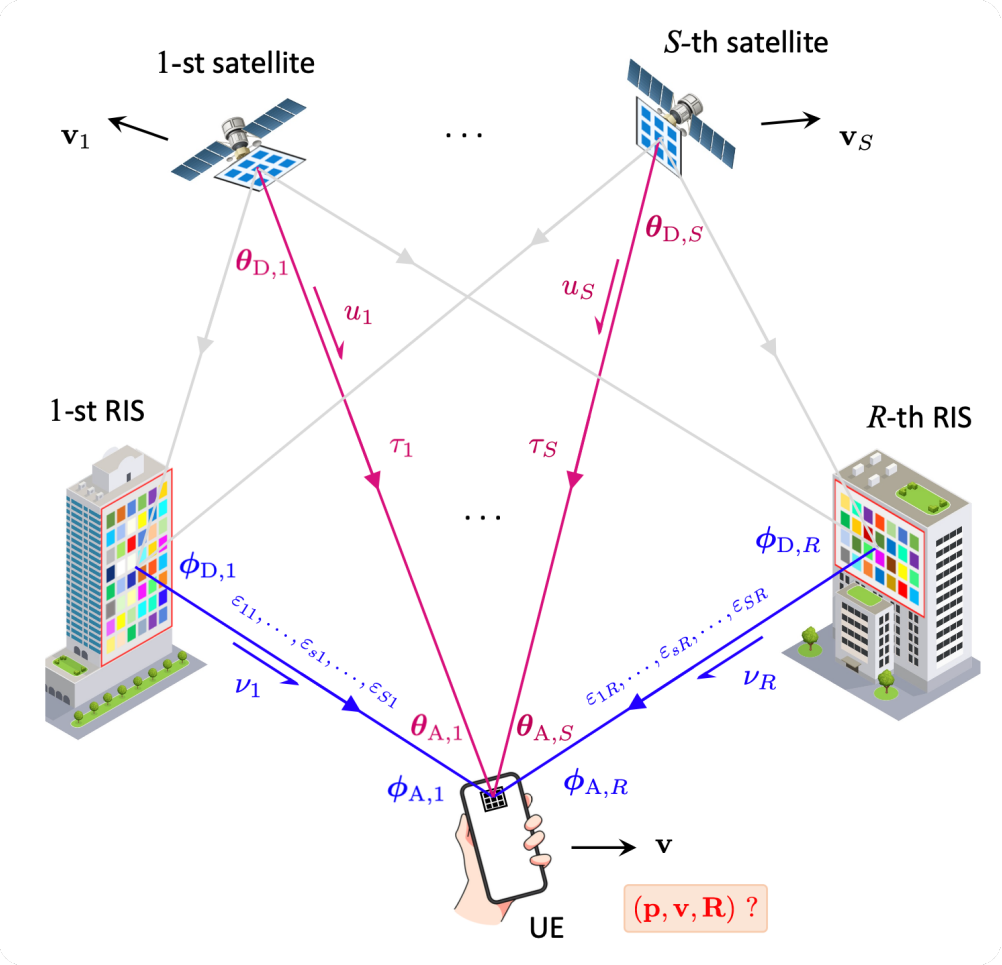
[4] S. Saleh et al., "6G RIS-aided Single-LEO Localization with Slow and Fast Doppler Effects," IEEE Globecom Workshops (GC Wkshps), Cape Town, South Africa, 2024, pp. 1-7,

# Case Study: RIS-Aided User Tracking

- LEO satellite signal frequency overlaps terrestrial network frequency bands.
- RIS can reflect LEO satellite signals [2].

**From snapshot localization to dynamic tracking:**

- 3D UE position
- 3D UE orientation
- 3D UE velocity





[1] P. Zheng, X. Liu, and T. Y. Al-Naffouri, "LEO- and RIS-Empowered User Tracking: A Riemannian Manifold Approach," in IEEE Journal of Selected Areas in Communications, vol. 42, no. 12, pp. 3445-3461, 2024.
[2] J. Ye, J. Qiao, A. Kammoun and M. -S. Alouini, "Nonterrestrial Communications Assisted by Reconfigurable Intelligent Surfaces," in Proceedings of the IEEE, vol. 110, no. 9, pp. 1423-1465, Sept. 2022.

# Case Study: RIS-Aided User Tracking

## A two-stage localization flow:

### Received signal

$$y_s^k = \sqrt{P_s}\mathbf{w}_s^{\mathsf{H}}\underline{\mathbf{H}}_s^k\mathbf{f}_s x_s^k + n_s^k,$$

$$\underline{\mathbf{H}}_s^k = \mathbf{H}_s^k + \sum_{r=1}^{R}\mathbf{H}_r^k\boldsymbol{\Gamma}_r\mathbf{H}_{sr}^k,$$

$$\mathbf{H}_\chi^k = \sqrt{\frac{K_f}{K_f+1}}\bar{\mathbf{H}}_\chi^k + \sqrt{\frac{1}{K_f+1}}\tilde{\mathbf{H}}_\chi^k,$$

Estimate

### Channel parameters

$$\boldsymbol{\eta}_0 = \left[\boldsymbol{\nu}^{\mathsf{T}}, \boldsymbol{\phi}_{\mathrm{D}}^{\mathsf{T}}, \boldsymbol{\phi}_{\mathrm{A}}^{\mathsf{T}}\right]^{\mathsf{T}} \in \mathbb{R}^{5R},$$

$$\boldsymbol{\eta}_s = \left[u_s, \boldsymbol{\theta}_{\mathrm{D},s}^{\mathsf{T}}, \boldsymbol{\theta}_{\mathrm{A},s}^{\mathsf{T}}, \tau_s, \varepsilon_{s1}, \ldots, \varepsilon_{sr}, \ldots, \varepsilon_{sR}\right]^{\mathsf{T}} \in \mathbb{R}^{R+6},$$

$$\boldsymbol{\nu} = \left[\nu_1, \ldots, \nu_r, \ldots, \nu_R\right]^{\mathsf{T}} \in \mathbb{R}^{R},$$

$$\boldsymbol{\phi}_{\mathrm{D}} = \left[\boldsymbol{\phi}_{\mathrm{D},1}^{\mathsf{T}}, \ldots, \boldsymbol{\phi}_{\mathrm{D},r}^{\mathsf{T}}, \ldots, \boldsymbol{\phi}_{\mathrm{D},R}^{\mathsf{T}}\right]^{\mathsf{T}} \in \mathbb{R}^{2R},$$

$$\boldsymbol{\phi}_{\mathrm{A}} = \left[\boldsymbol{\phi}_{\mathrm{A},1}^{\mathsf{T}}, \ldots, \boldsymbol{\phi}_{\mathrm{A},r}^{\mathsf{T}}, \ldots, \boldsymbol{\phi}_{\mathrm{A},R}^{\mathsf{T}}\right]^{\mathsf{T}} \in \mathbb{R}^{2R}.$$

Estimate

### Localization parameters

$$\mathbf{r} = [\mathbf{p}^{\mathsf{T}}, \mathbf{v}^{\mathsf{T}}, \mathbf{b}^{\mathsf{T}}, \mathrm{vec}(\mathbf{R})^{\mathsf{T}}]^{\mathsf{T}} \in \mathbb{R}^{S+15}$$

pos. vel. clock bias ori.

Assume an **efficient channel estimator** is applied,
so the estimation errors is characterized by **CRB**.

# Case Study: RIS-Aided User Tracking

## The two-stage localization flow:

Channel parameters

$$\boldsymbol{\eta}_0 = [\boldsymbol{\nu}^\mathsf{T}, \boldsymbol{\phi}_\mathrm{D}^\mathsf{T}, \boldsymbol{\phi}_\mathrm{A}^\mathsf{T}]^\mathsf{T} \in \mathbb{R}^{5R},$$
$$\boldsymbol{\eta}_s = [u_s, \boldsymbol{\theta}_{\mathrm{D},s}^\mathsf{T}, \boldsymbol{\theta}_{\mathrm{A},s}^\mathsf{T}, \tau_s, \varepsilon_{s1}, \ldots, \varepsilon_{sr}, \ldots, \varepsilon_{sR}]^\mathsf{T} \in \mathbb{R}^{R+6},$$
$$\boldsymbol{\nu} = [\nu_1, \ldots, \nu_r, \ldots, \nu_R]^\mathsf{T} \in \mathbb{R}^R,$$
$$\boldsymbol{\phi}_\mathrm{D} = [\boldsymbol{\phi}_{\mathrm{D},1}^\mathsf{T}, \ldots, \boldsymbol{\phi}_{\mathrm{D},r}^\mathsf{T}, \ldots, \boldsymbol{\phi}_{\mathrm{D},R}^\mathsf{T}]^\mathsf{T} \in \mathbb{R}^{2R},$$
$$\boldsymbol{\phi}_\mathrm{A} = [\boldsymbol{\phi}_{\mathrm{A},1}^\mathsf{T}, \ldots, \boldsymbol{\phi}_{\mathrm{A},r}^\mathsf{T}, \ldots, \boldsymbol{\phi}_{\mathrm{A},R}^\mathsf{T}]^\mathsf{T} \in \mathbb{R}^{2R}.$$

Estimate →

Localization parameters

$$\mathbf{r} = [\mathbf{p}^\mathsf{T}, \mathbf{v}^\mathsf{T}, \mathbf{b}^\mathsf{T}, \mathrm{vec}(\mathbf{R})^\mathsf{T}]^\mathsf{T} \in \mathbb{R}^{S+15}$$

pos. vel. clock bias ori.

**Observations** ↑ ↑ **Unknowns**

How to estimate in a **recursive** way? ⇒ Unscented Kalman filter (UKF)

## Nonlinear state-space model in UKF:

$n$: index of the time instant.

$$Time\ update: \ \mathbf{r}_{n+1} = f(\mathbf{r}_n; \hat{\mathbf{a}}_n, \hat{\boldsymbol{\Omega}}_n) + \Delta\mathbf{r}_{n+1},$$
$$Measurement\ update: \ \hat{\boldsymbol{\eta}}_n = h(\mathbf{r}_n) + \Delta\boldsymbol{\eta}_n.$$

Process function

Observation function

UE **acceleration** and **rotation**, assumed can be acquired through an inertial measurement unit (IMU).

[1] P. Zheng, X. Liu, and T. Y. Al-Naffouri, "LEO- and RIS-Empowered User Tracking: A Riemannian Manifold Approach," in IEEE Journal of Selected Areas in Communications, vol. 42, no. 12, pp. 3445-3461, 2024.

# Case Study: RIS-Aided User Tracking

**Nonlinear state-space model in UKF:**

$n$: index of the time instant.

$$Time\ update:\ \mathbf{r}_{n+1} = f(\mathbf{r}_n; \hat{\mathbf{a}}_n, \hat{\mathbf{\Omega}}_n) + \Delta\mathbf{r}_{n+1},$$
$$Measurement\ update:\ \hat{\boldsymbol{\eta}}_n = h(\mathbf{r}_n) + \Delta\boldsymbol{\eta}_n.$$

Process function — Observation function

$$\mathbf{p}_{n+1} = \mathbf{p}_n + \mathbf{v}_n\delta_t + \bar{\mathbf{a}}_n\frac{\delta_t^2}{2},$$
$$\mathbf{v}_{n+1} = \mathbf{v}_n + \bar{\mathbf{a}}_n\delta_t,$$
$$\mathbf{b}_{n+1} = \mathbf{b}_n,$$
$$\mathbf{R}_{n+1} = \mathbf{R}_n\bar{\mathbf{\Omega}}_n,$$

**The Satellite-UE Direct Channel:**

Doppler shift: $$u_s = \frac{(\mathbf{v}_s - \mathbf{v})^\mathsf{T}(\mathbf{p} - \mathbf{p}_s)}{\lambda_c\|\mathbf{p} - \mathbf{p}_s\|_2}.$$

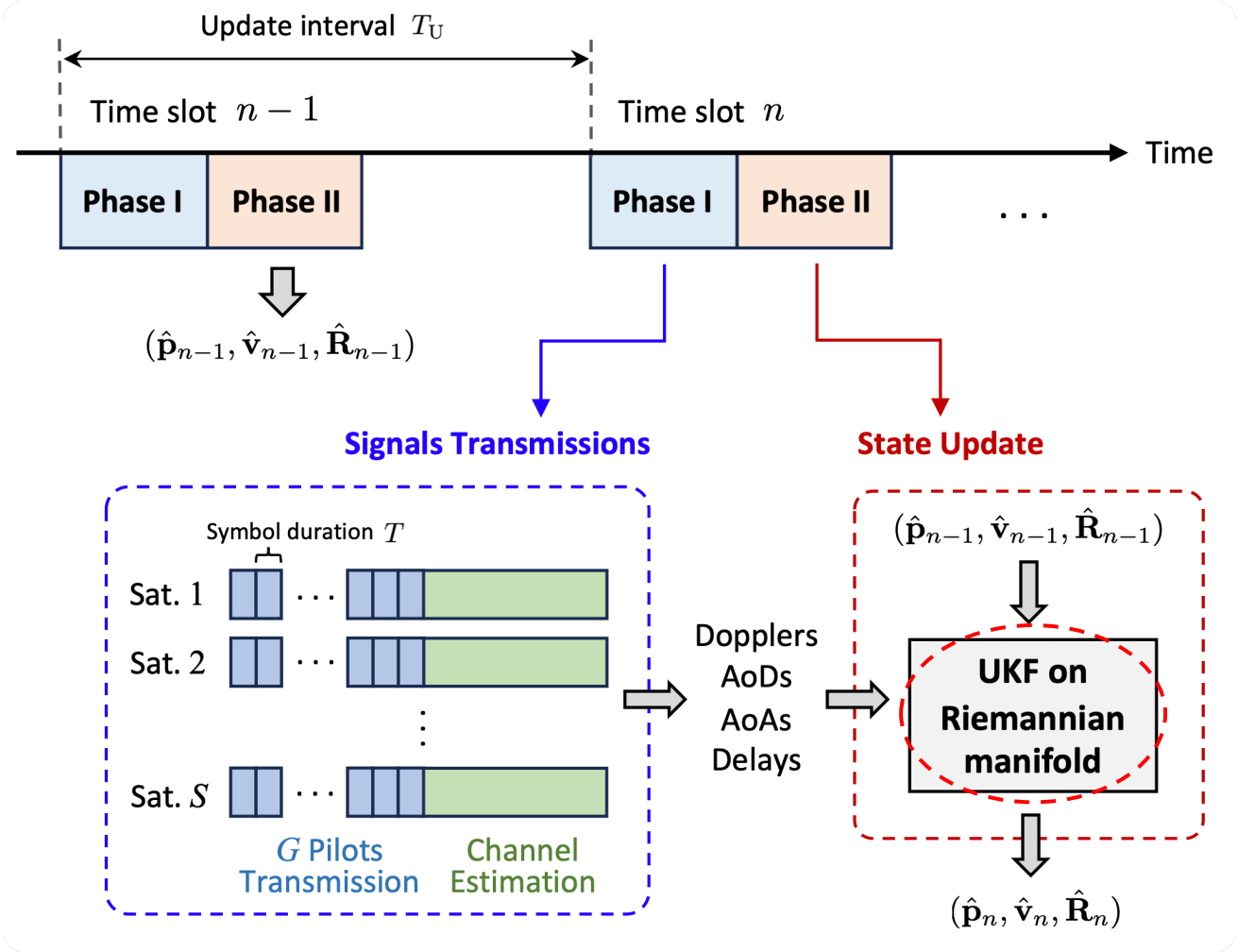
AoD at satellite:
$$\theta_{\mathrm{D},s}^{\mathrm{az}} = \mathrm{atan2}\big([\mathbf{R}_s^\mathsf{T}(\mathbf{p} - \mathbf{p}_s)]_2, [\mathbf{R}_s^\mathsf{T}(\mathbf{p} - \mathbf{p}_s)]_1\big),$$
$$\theta_{\mathrm{D},s}^{\mathrm{el}} = \mathrm{asin}\big([\mathbf{R}_s^\mathsf{T}(\mathbf{p} - \mathbf{p}_s)]_3/\|\mathbf{p} - \mathbf{p}_s\|_2\big).$$

AoA at UE:
$$\theta_{\mathrm{A},s}^{\mathrm{az}} = \mathrm{atan2}\big([\mathbf{R}^\mathsf{T}(\mathbf{p}_s - \mathbf{p})]_2, [\mathbf{R}^\mathsf{T}(\mathbf{p}_s - \mathbf{p})]_1\big),$$
$$\theta_{\mathrm{A},s}^{\mathrm{el}} = \mathrm{asin}\big([\mathbf{R}^\mathsf{T}(\mathbf{p}_s - \mathbf{p})]_3/\|\mathbf{p}_s - \mathbf{p}\|_2\big).$$

Channel delay: $$\tau_s = \frac{\|\mathbf{p}_s - \mathbf{p}\|_2}{c} + b_s,$$

[1] P. Zheng, X. Liu, and T. Y. Al-Naffouri, "LEO- and RIS-Empowered User Tracking: A Riemannian Manifold Approach," in IEEE Journal of Selected Areas in Communications, vol. 42, no. 12, pp. 3445-3461, 2024.

# Case Study: RIS-Aided User Tracking




**Main Challenges:**

- Constrained localization space
- Unknown obs. statistics

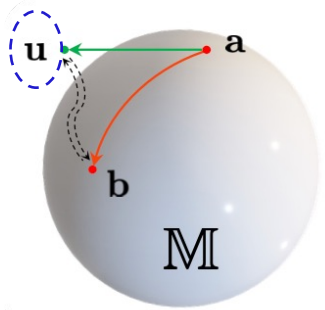
# Case Study: RIS-Aided User Tracking

- **Challenge 1:** Constrained localization space

$$\mathbf{p} \in \mathbb{R}^3,\ \mathbf{v} \in \mathbb{R}^3,\ \mathbf{b} \in \mathbb{R}^S, \mathbf{R} \in \mathrm{SO}(3).$$

$$\mathrm{SO}(3) \triangleq \{\mathbf{R} | \mathbf{R}^\mathsf{T}\mathbf{R} = \mathbf{I}_3, \det(\mathbf{R}) = 1\}.$$

Euclidean space




$$\mathbf{a} \boxplus \mathbf{u} = \mathbf{b}$$
$$\mathbf{b} \boxminus \mathbf{a} = \mathbf{u}$$

Projection onto Riemannian manifold → Defined Riemannian manifold:

$$\mathbb{M} \triangleq \mathbb{R}^3 \times \mathbb{R}^3 \times \mathbb{R}^S \times \mathrm{SO}(3)$$

- **Challenge 2:** Unknown obs. statistics

The success of Kalman filter relies on the knowledge of the **statistics** of the observations.

**Covariance matrix** can be characterized by **Fisher information matrix (FIM)**.

However, computing FIM need the **true UE state**, which is unknown. → **Can we guess it?**

# Case Study: RIS-Aided User Tracking

➢ **Challenge 2:** Unknown obs. statistics

Time $n$-1

UE state: $\mathbf{r}_{n-1}$

Cov. matrix: $\mathbf{P}_{n-1}$

Generate sigma points

$$\mathbf{z}_0 = \bar{\mathbf{r}}_{n-1},$$
$$\mathbf{z}_i = \bar{\mathbf{r}}_{n-1} \boxplus \left(\sqrt{(N_\mathrm{p}+\Lambda)\mathbf{P}_{n-1}}\right)_i, \ i = 1, \ldots, N_\mathrm{p},$$
$$\mathbf{z}_{i+N_\mathrm{p}} = \bar{\mathbf{r}}_{n-1} \boxplus \left(\sqrt{(N_\mathrm{p}+\Lambda)\mathbf{P}_{n-1}}\right)_i, \ i = 1, \ldots, N_\mathrm{p},$$

Predict time $n$ based on the process function $f(\cdot)$

$$\tilde{\mathbf{r}}_n = \mathrm{MoSP}\left(\{W_i^{(m)}\}, \{f(\mathbf{z}_i)\}\right),$$
$$\tilde{\mathbf{P}}_n = \sum_{i=0}^{2N_\mathrm{p}} W_i^{(c)} \left(f(\mathbf{z}_i) \boxminus \tilde{\mathbf{r}}_n\right)\left(f(\mathbf{z}_i) \boxminus \tilde{\mathbf{r}}_n\right)^\mathsf{T} + \mathbf{P},$$

Generate sigma points

$$\mathbf{x}_0 = \tilde{\mathbf{r}}_n,$$
$$\mathbf{x}_i = \tilde{\mathbf{r}}_n \boxplus \left(\sqrt{(N_\mathrm{p}+\Lambda)\tilde{\mathbf{P}}_n}\right)_i, \ i = 1, \ldots, N_\mathrm{p},$$
$$\mathbf{x}_{i+N_\mathrm{p}} = \tilde{\mathbf{r}}_n \boxplus \left(\sqrt{(N_\mathrm{p}+\Lambda)\tilde{\mathbf{P}}_n}\right)_i, \ i = 1, \ldots, N_\mathrm{p}.$$

Can be used to compute Fisher information matrix (FIM) to approximate the obs. covariance!

Correct prediction based on the observation at time $n$

$$\tilde{\boldsymbol{\eta}}_n = \sum_{i=0}^{2N_\mathrm{p}} W_i^{(m)} h(\mathbf{x}_i),$$
$$\tilde{\mathbf{Q}}_n = \sum_{i=0}^{2N_\mathrm{p}} W_i^{(c)} \left(h(\mathbf{x}_i) - \tilde{\boldsymbol{\eta}}_n\right)\left(h(\mathbf{x}_i) - \tilde{\boldsymbol{\eta}}_n\right)^\mathsf{T} + \boldsymbol{\Sigma}_n.$$
$$\tilde{\mathbf{Q}}_{r\eta,n} = \sum_{i=0}^{2N_\mathrm{p}} W_i^{(c)} \left(f(\mathbf{z}_i) \boxminus \tilde{\mathbf{r}}_n\right)\left(h(\mathbf{x}_i) - \tilde{\boldsymbol{\eta}}_n\right)^\mathsf{T}$$
$$\mathbf{K}_n = \tilde{\mathbf{Q}}_{r\eta,n} \tilde{\mathbf{Q}}_n^{-1},$$

*Need to know the Cov. matrix of obs.*

Estimate at time $n$

$$\hat{\mathbf{r}}_n = \tilde{\mathbf{r}}_n \boxplus \mathbf{K}_n(\hat{\boldsymbol{\eta}}_n - \tilde{\boldsymbol{\eta}}_n),$$
$$\hat{\mathbf{P}}_n = \tilde{\mathbf{P}}_n - \mathbf{K}_n \tilde{\mathbf{R}}_n \mathbf{K}_n^\mathsf{T}.$$

# Case Study: RIS-Aided User Tracking

- **Challenge 2:** Unknown obs. statistics

Time $n$-1 → Generate sigma points → Predict time $n$ based on the process function $f(\cdot)$

Generate sigma points

$$\begin{aligned}\mathbf{x}_0 &= \tilde{\mathbf{r}}_n,\\ \mathbf{x}_i &= \tilde{\mathbf{r}}_n \boxplus \left(\sqrt{(N_\mathrm{p}+\Lambda)\tilde{\mathbf{P}}_n}\right)_i,\ i=1,\ldots,N_\mathrm{p},\\ \mathbf{x}_{i+N_\mathrm{p}} &= \tilde{\mathbf{r}}_n \boxplus \left(\sqrt{(N_\mathrm{p}+\Lambda)\tilde{\mathbf{P}}_n}\right)_i,\ i=1,\ldots,N_\mathrm{p}.\end{aligned}$$

Correct prediction based on the observation at time $n$

$$\begin{aligned}\tilde{\boldsymbol{\eta}}_n &= \sum_{i=0}^{2N_\mathrm{p}} W_i^{(m)} h(\mathbf{x}_i),\\ \tilde{\mathbf{Q}}_n &= \sum_{i=0}^{2N_\mathrm{p}} W_i^{(c)} \left(h(\mathbf{x}_i)-\tilde{\boldsymbol{\eta}}_n\right)\left(h(\mathbf{x}_i)-\tilde{\boldsymbol{\eta}}_n\right)^\mathsf{T} + \boldsymbol{\Sigma}_n,\\ \tilde{\mathbf{Q}}_{r\eta,n} &= \sum_{i=0}^{2N_\mathrm{p}} W_i^{(c)} \left(f(\mathbf{z}_i) \boxminus \tilde{\mathbf{r}}_n\right)\left(h(\mathbf{x}_i)-\tilde{\boldsymbol{\eta}}_n\right)^\mathsf{T},\\ \mathbf{K}_n &= \tilde{\mathbf{Q}}_{r\eta,n}\tilde{\mathbf{Q}}_n^{-1},\end{aligned}$$

Estimate at time $n$

$$\begin{aligned}\hat{\mathbf{r}}_n &= \tilde{\mathbf{r}}_n \boxplus \mathbf{K}_n(\hat{\boldsymbol{\eta}}_n - \tilde{\boldsymbol{\eta}}_n),\\ \hat{\mathbf{P}}_n &= \tilde{\mathbf{P}}_n - \mathbf{K}_n\tilde{\mathbf{R}}_n\mathbf{K}_n^\mathsf{T}.\end{aligned}$$

Proposed ***belief assignment principle*** →

$$\hat{\boldsymbol{\Sigma}}_n = \epsilon\mathbf{I} + \sum_{i=0}^{2N_\mathrm{P}} W_i^{(c)} \mathbf{J}^{-1}(\mathbf{x}_i)$$

Regularization term | Weights | FIM computed based on each sigma point

# Case Study: RIS-Aided User Tracking

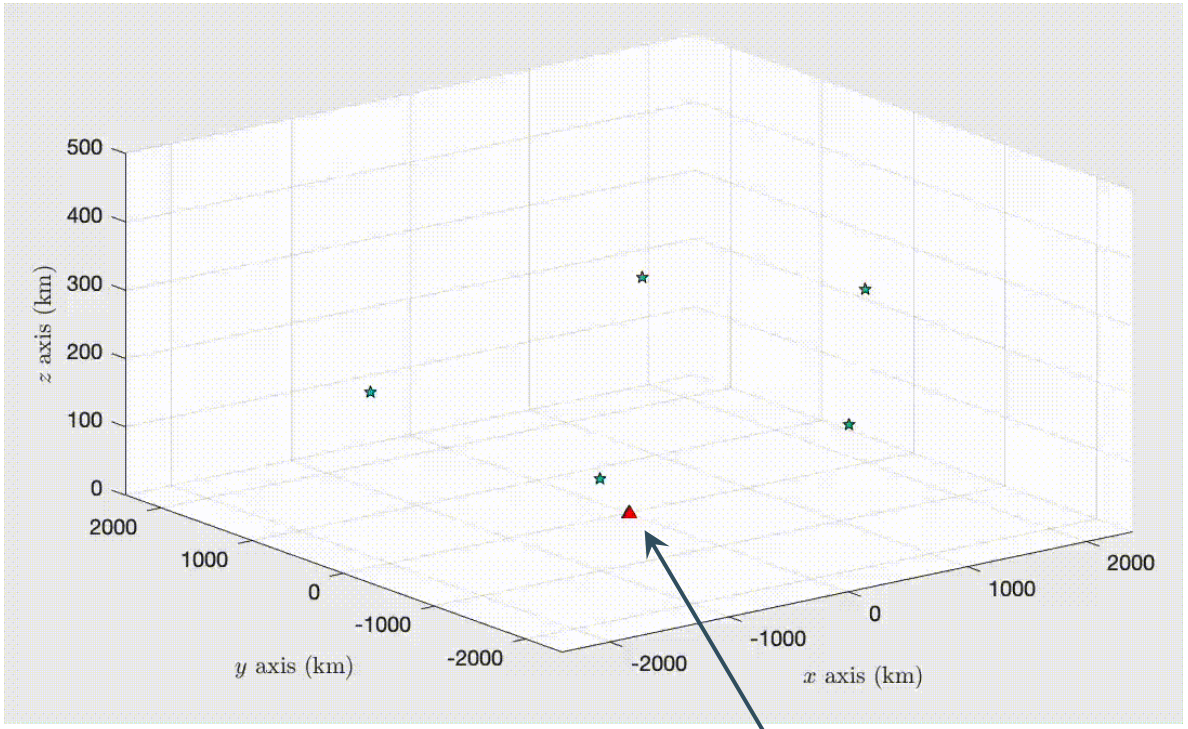
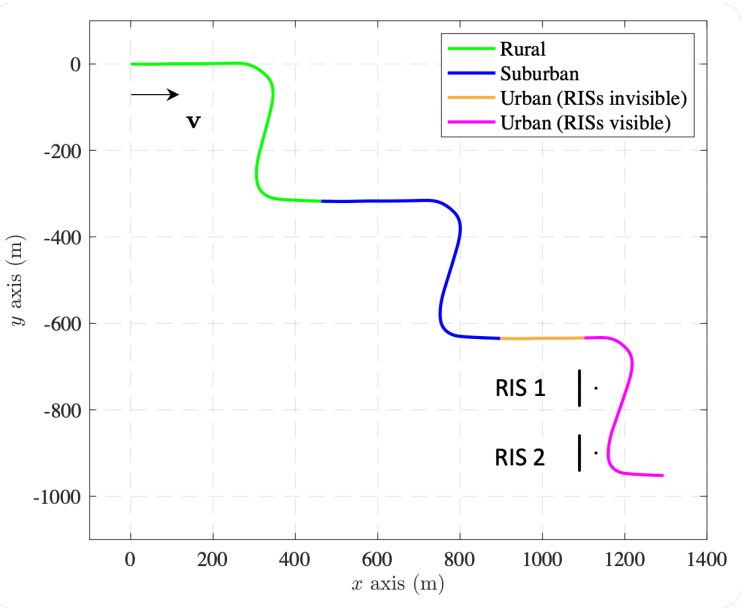



Table 6.2: Default Simulation Parameters

| Parameter | Value |
|---|---|
| Number of LEO satellites $S$ | 5 |
| Number of RISs $R$ | 2 |
| Number of Subcarriers $K$ | 3000 |
| Number of Transmissions $G$ | 32 |
| Carrier Frequency $f_c$ | 12.7 GHz |
| Bandwidth $B$ | 240 MHz |
| Transmission Power $P_{\mathrm{T}}$ | 50 dBm |
| Active RIS Power $P_{\mathrm{R}}$ | 0 dBm |
| Noise PSD of Receiver & RIS | $-174$ dBm/Hz |
| Noise Figure of Receiver & RIS | 0 dB |
| Array Size of satellite/RIS/UE | $4 \times 4/20 \times 20/4 \times 4$ |
| Array Spacing of satellite/RIS/UE | 5 mm/5 mm/5 mm |
| Rural/Suburban/Urban $K_f$ [150] | 3 dB/2.6 dB/1.85 dB |
| Acceleration standard deviation $\sigma_a$ | $0.2\,\mathrm{m/s^2}$ |
| Rotation standard deviation $\sigma_\omega$ | $2^\circ$ |
| Update Interval $T_{\mathrm{U}}$ | 1 s |
| Orbit Height of LEO satellites | 500 km |
| Antenna correlation $\boldsymbol{\Theta}/\boldsymbol{\Theta}_s/\boldsymbol{\Theta}_r$ | $\mathbf{I}$ |

[1] P. Zheng, X. Liu, and T. Y. Al-Naffouri, "LEO- and RIS-Empowered User Tracking: A Riemannian Manifold Approach," in IEEE Journal of Selected Areas in Communications, vol. 42, no. 12, pp. 3445-3461, 2024.

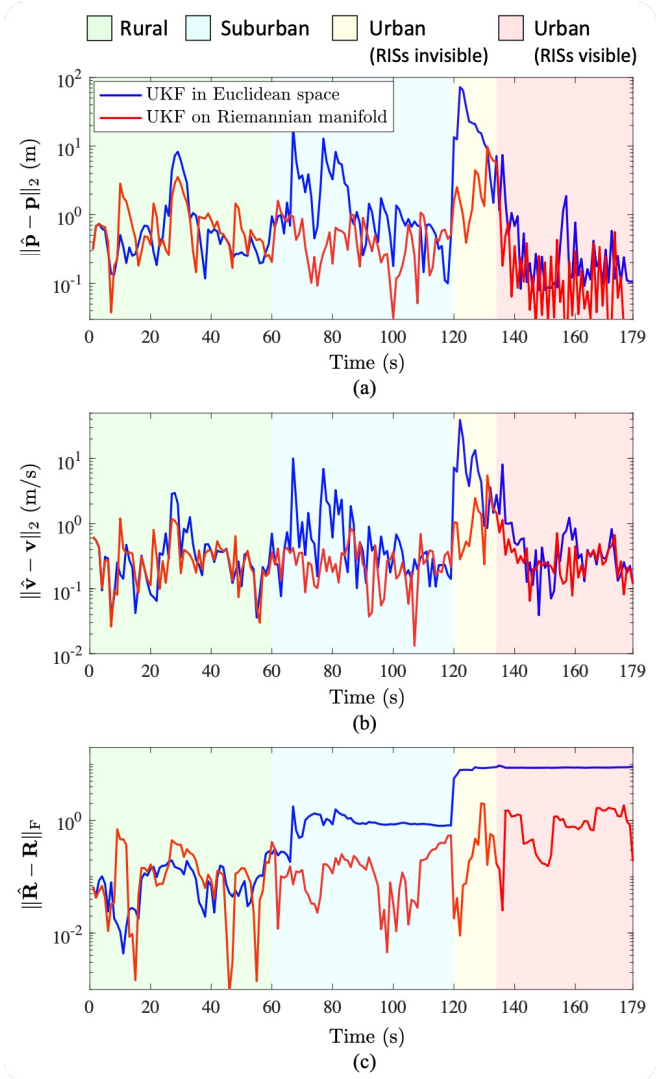
# Case Study: RIS-Aided User Tracking




$$\mathbf{H}_{\chi}^{k} = \sqrt{\frac{K_f}{K_f+1}}\bar{\mathbf{H}}_{\chi}^{k} + \sqrt{\frac{1}{K_f+1}}\tilde{\mathbf{H}}_{\chi}^{k}$$

Rician factor $K_f$:

Rural/suburban/urban: 3/2.6/1.85 dB




- ✓ Without the presence of RISs, the tracking performance of the system exhibits a trend:

  *Rural > Suburban > Urban*

- ✓ RIS can enhance tracking performance of UE position and velocity in urban regions.

- ✓ Comparing the UKF in Euclidean space and on Riemannian manifold:
  - In **rural regions**: similar accuracy
  - In **suburban and urban regions**: the proposed Riemannian manifold-based UKF can consistently outperform the classical Euclidean UKF.

[1] P. Zheng, X. Liu, and T. Y. Al-Naffouri, "LEO- and RIS-Empowered User Tracking: A Riemannian Manifold Approach," in IEEE Journal of Selected Areas in Communications, vol. 42, no. 12, pp. 3445-3461, 2024.

# Key Takeaways

- **LEO vs. GNSS**: LEO satellites serve as an important complement to GNSS
- **LEO signal and observation model**: Highly heterogeneous, depending on system design
- **New challenges**: Orbital errors, frequent handover, signal processing, …
- **New opportunities**: Integrated positioning and communication, and integrated TN and NTN localization, more advanced signal processing approaches are needed!